Project #18: Madigan CDC Back Up Power Building 6995

Project Title: Madigan Back-Up Generator

Project Scope:

Building 6995 is on the JBG Critical Asset List and therefore authorized to have an emergency backup generator. The building requires backup power to provide uninterrupted power to support sustained mission operations during emergency operations.

Building 6995 has a 1600Amp 208Y/120 service.

Provide and install a new diesel-powered emergency generator providing stand-by power support to the whole building with a minimum set rating: 500kW, 208Y/120V, 3PH

The new generator shall have an integrated sub-base fuel tank incorporating double-wall Washington State environmentally compliant construction. The fuel capacity for the sub-base fuel tank shall provide a minimum 'run-time' of 48-continuous hours and incorporating 'Energy Efficient' Block Heaters and site gauge as sub-base fuel tank.

Generator controller shall be capable to interface the generator set to Enterprise Energy Data Reporting System (EEDRS) equipment using Modbus communications protocols in accordance with JBLM 26 27 13.10 30 Rev 11, Para 3.3.8. Generators must have a remote emergency stop switch to shut down the prime mover, and it shall be installed on the exterior of the generator enclosure per NEC Section 445.19(B). The new Generator should include electrical packages inside of the generator which included GFCI convenience receptacles and lights.

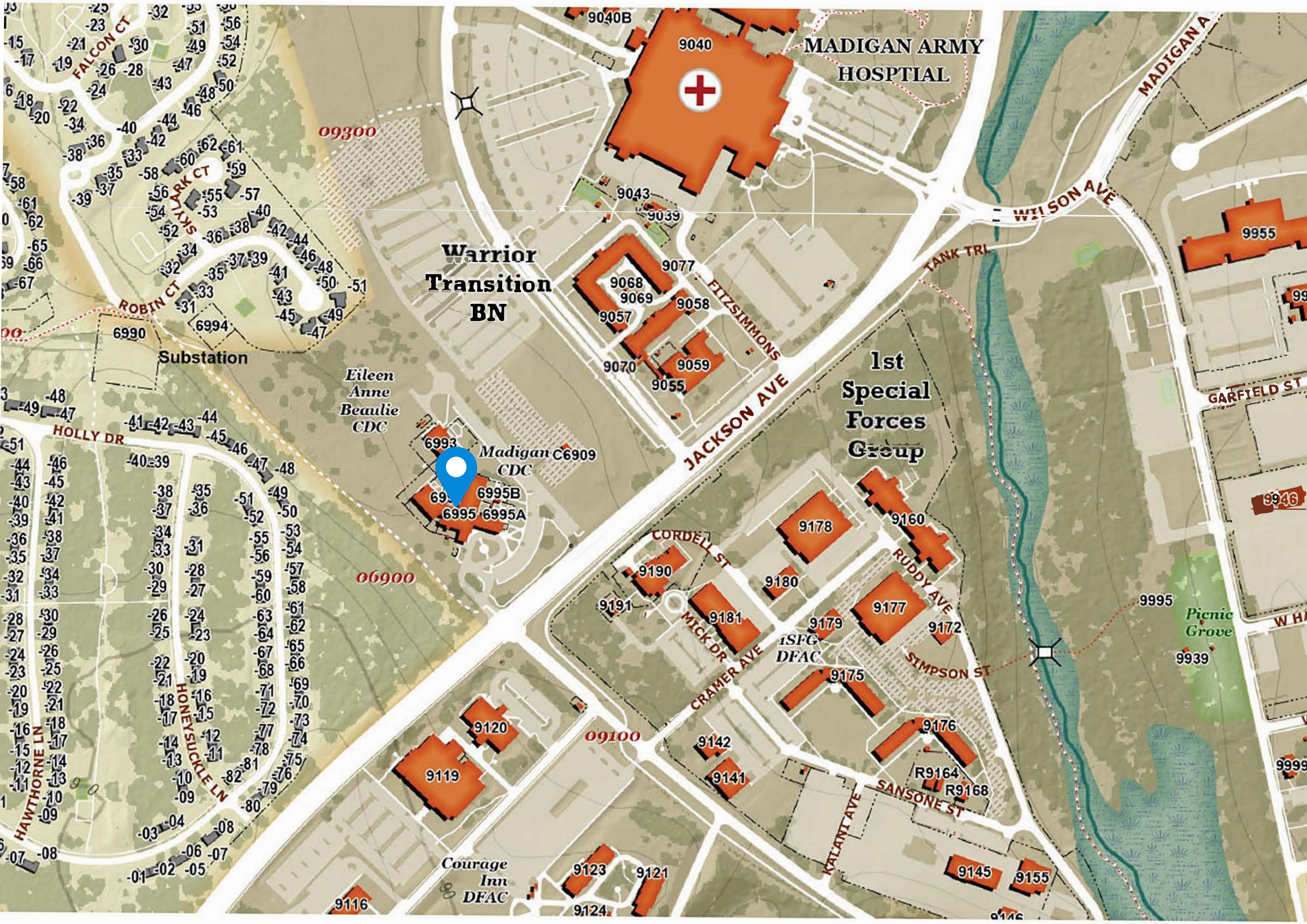
Provide and install a 3-way manual transfer switch, 1600A, 208Y/120V to allow safe isolated load banking testing and connection of a portable backup generator tied in at the same time. (NEMA 3R)

Provide and install a new 1600A, 208Y/120V ATS. The new ATS shall have isolation bypass with 4 pole configuration. (NEMA 3R)

Provide arc flash study for newly installed equipment.

The advanced metering system - Enterprise Energy Data Reporting System (EEDRS) and JACE require replacement to interconnect with the generator set. The intent of this is to provide the metering data and generator data to the local engineering staff first within the building automation system that can execute application programs to affect savings and able track energy usage. Electricity derived from a generator is charged at a different rate than utility provided power. EEDRS is used to monitor generator output. It allows them to ensure the generator is performing properly and not producing unsatisfactory power that could cause damage to other building systems. The metering system in Building 6995 is in failing condition due to corrosion, power surges, lack of preventative maintenance, and general failures as display non functional, date and time in the meter get drifted, and clearance provided between name plate and the casing of the meter was not enough that create an unsafe system. Additionally, it can no longer be calibrated due to age. EEDRS JACE is a requirement per UFC 3-400-01 and JBLM Design Standards to be code-compliant with the installation of metering.

All work shall comply with the most current editions of NFPA 110, 70, 70E, and JBLM Design Standard



MADIGAN ARMY HOSPTIAL
9040
9040B
9043
9039
Warrior Transition BN
9077
9068
9069
9058
9057
9070
9059
9055
FITZSIMMONS
JACKSON AVE
WILSON AVE
MADIGAN A
TANK TRL
9955
GARFIELD ST
1st Special Forces Group
9178
9160
CORDELL ST
9190
9180
RUDDY AVE
9177
9172
9191
9181
MICK DR
9179
1SFG DFAC
SIMPSON ST
9175
CRAMER AVE
9176
9142
9141
R9164
R9168
SANSONE ST
KALANI AVE
9145
9155
9995
Picnic Grove
9939
9946
9999
W HA
09300
09100
06900
Eileen Anne Beaulie CDC
6993
Madigan CDC
C6909
6995B
6995
6995A
6990
Substation
6994
ROBIN CT
SKYLARK CT
FALCON CT
HOLLY DR
HAWTHORNE LN
HONEYSUCKLE LN
9120
9119
Courage Inn DFAC
9123
9121
9116
9124

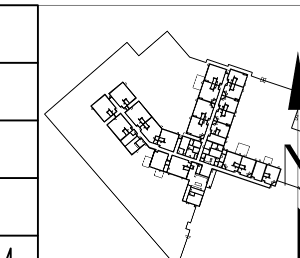
BLDG 6995
1st FLOOR
FAMILY SUPP (CHILD DEVELOPMENT)

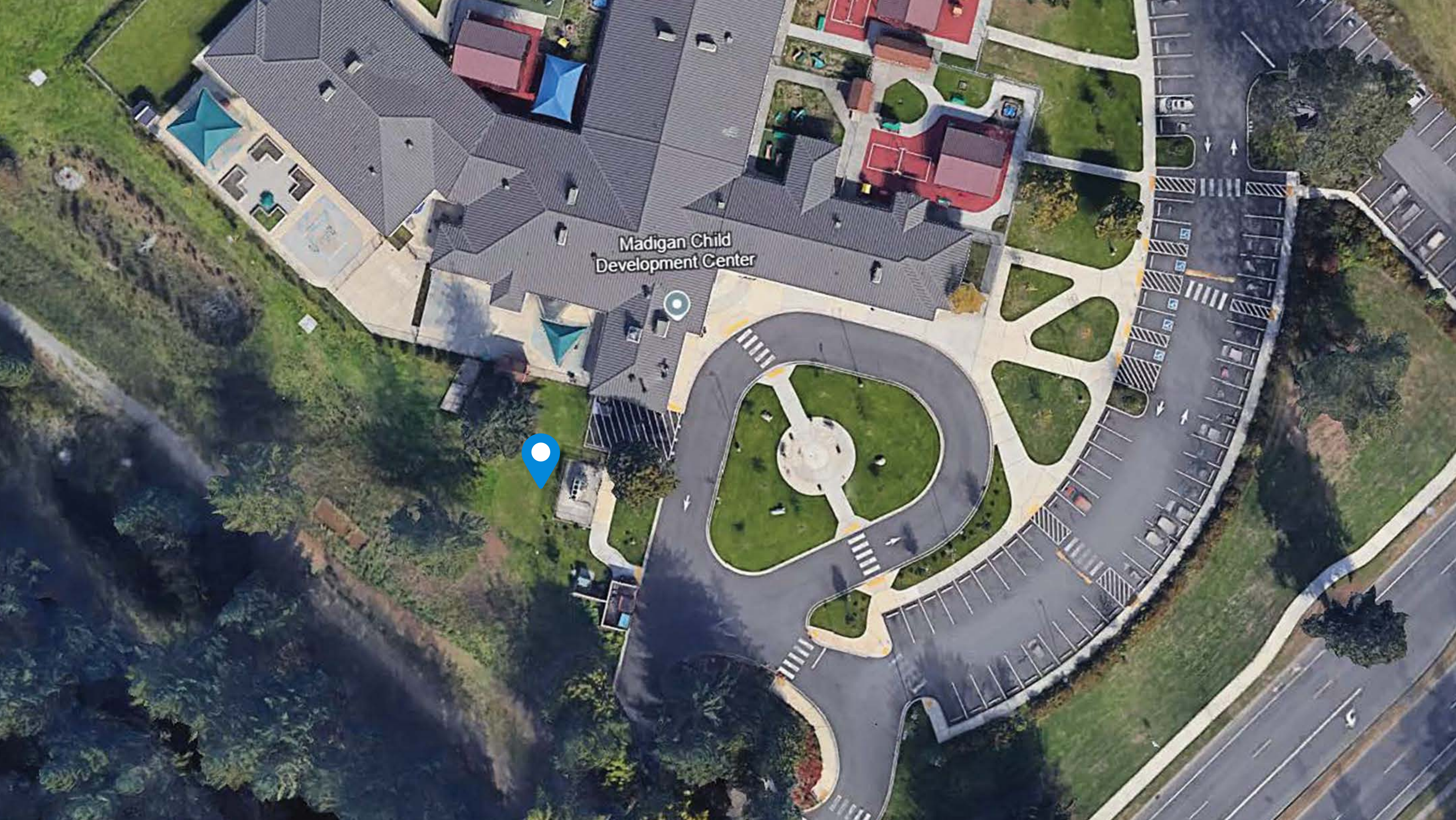
FLOOR PLAN

SURVEY DATE: 6 APRIL 2015

NOT TO SCALE

31 August 2016 8:23:59 AM



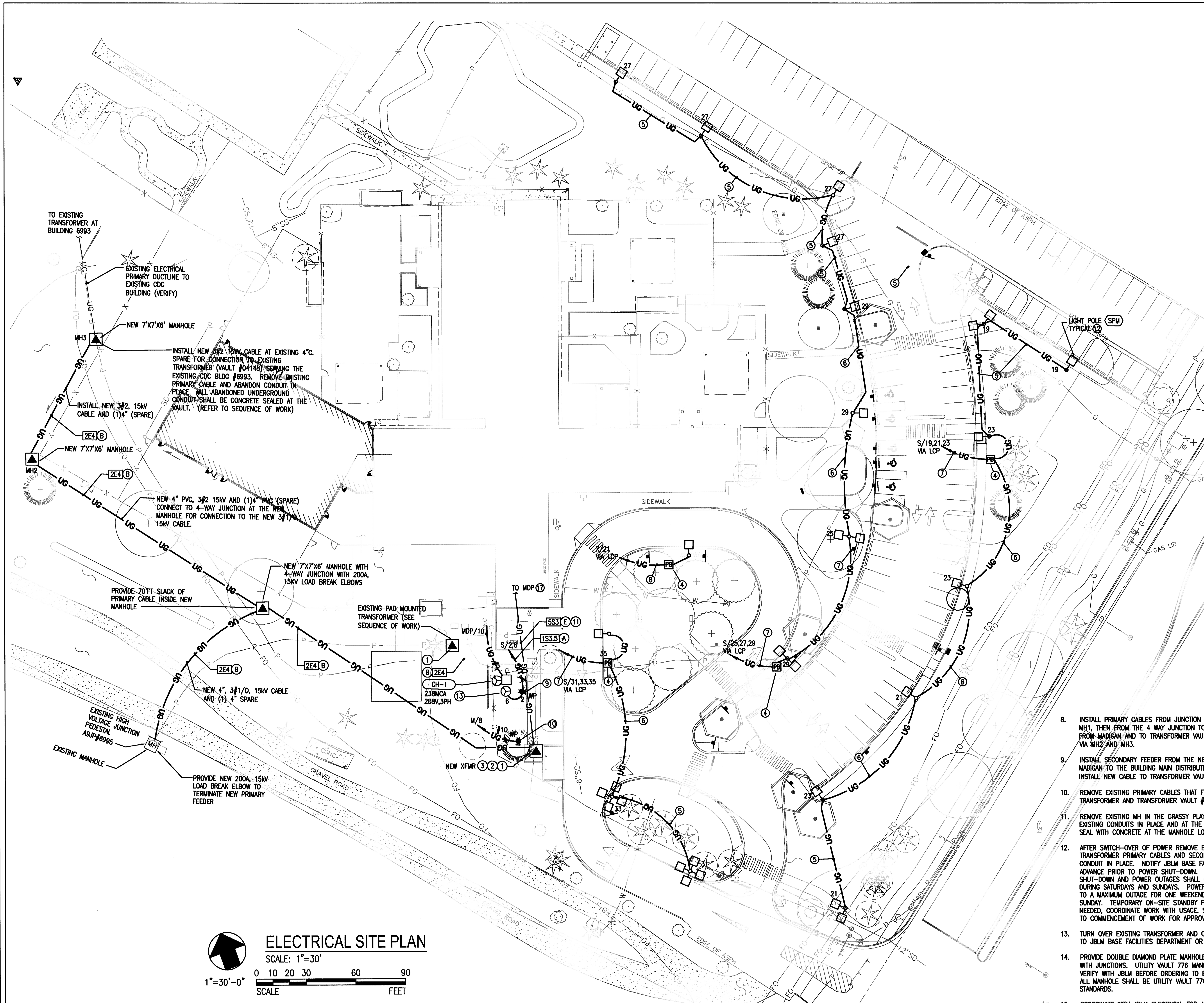

Madigan Child
Development Center

PUBLIC WORKS

PROJECT TITLE:
PROPOSED GENERATOR LOCATION

DRAWING TITLE:
BUILDING 6995

WORK ORDER:		POC:
–		–
DATE: 10/03/2024	BUILDING NO.: R9642	FILE NO.: 0000-01.DWG



ELECTRICAL SITE PLAN

SCALE: 1"=30'

1"=30'-0" 0 10 20 30 60 90 SCALE FEET

GENERAL NOTES:

1. COORDINATE LOCATION OF POLES WITH LANDSCAPE PLANS TO MAINTAIN PERIMETER ROOT PROTECTION OF EXISTING AND NEW TREES. POLE LIGHTS CIRCUIT ROUTING SHALL BE VERIFIED AND COORDINATED IN FIELD TO NOT BE IN CONFLICT WITH THE EXISTING AND NEW TREES. PROVIDE ADDITIONAL PULLBOX AS REQUIRED PER CODE.
2. MAINTAIN MINIMUM SPACING OF 12" BETWEEN POWER AND COMMUNICATION LINES ON ALL UNDERGROUND DUCT SYSTEMS.

SHEET NOTES:

1. REDUCED TRANSFORMER SIZE TO 300KVA AS PER DIRECTION FROM JBLM BASE FACILITY DEPARTMENT.
2. PROVIDE CONCRETE TRANSFORMER VAULT.
3. PROVIDE NEW 300KVA TRANSFORMER. 13.8KV–208Y/120V,3PH,4W, 5.75%Z.
4. INSTALL 13"X24" PULLBOX FOR POLE LIGHT CIRCUIT ROUTING. REFER TO DETAIL 4/E502
5. 1"C–2#6 & 1#10 G IN PVC SCHEDULE 80
6. 1"C–3#6 & 1#10 G IN PVC SCHEDULE 80
7. 1"C–4#6 & 1#10 G IN PVC SCHEDULE 80
8. 1"C–2#8 & 1#12 G IN PVC SCHEDULE 80
9. MOUNT ON SIDE OF CHILLER EQUIPMENT ENCLOSURE.
10. IRRIGATION CONTROLLER, COORDINATE EXACT LOCATION AND REQUIREMENTS WITH LANDSCAPE ARCHITECT.
11. SAW CUT FLOOR, IN MECHANICAL ROOM, TO ROUTE FEEDERS FROM TRANSFORMER TO MAIN DISTRIBUTION PANEL. COORDINATE WORK WITH STRUCTURAL DRAWING FOR SAW CUTTING OF EXISTING SLAB. REPAIR FLOOR TO MATCH EXISTING.
12. REFER TO DETAIL 6/E501 FOR TYPICAL POLE LIGHT MOUNTING ON CONCRETE POLE BASE.
13. PROVIDE 120V CONNECTION TO CH–1 HEAT TRACE. HEAT TRACE SYSTEM PROVIDED AS PART OF CH–1 UNIT. COORDINATE EXACT POINT OF CONNECTION AND REQUIREMENTS WITH MECHANICAL PRIOR TO ROUGH–IN.
14. NOT USED.
15. NOT USED.
16. NOT USED.
17. REFER TO STRUCTURAL DETAIL FOR INSTALLATION OF NEW (2) 4"C IN CONCRETE SLAB.

SEQUENCE OF WORK:

1. MAINTAIN, PROTECT AND KEEP OPERATIONAL THE EXISTING PRIMARY DUCTLINES AND EXISTING PAD MOUNTED TRANSFORMER PRIOR TO SWITCH–OVER TO NEW PRIMARY DISTRIBUTION SYSTEM AND THE INSTALLATION OF THE NEW TRANSFORMER. MINIMIZE DOWNTIME.
2. INSTALL NEW 7' X 7' MANHOLE (MH1) AT LOCATION SHOWN. ADJUST LOCATION TO AVOID EXISTING DUCTBANKS.
3. INSTALL NEW PRIMARY DUCTLINES FROM EXISTING HIGH VOLTAGE JUNCTION PEDESTAL #A9JP2–6995 TO THE MH1.
4. INSTALL NEW PRIMARY DUCTLINE FROM MH1 TO NEW VAULT MOUNTED TRANSFORMER SERVING THE MADIGAN BLDG.
5. INSTALL NEW PAD–MOUNTED TRANSFORMER ON THE TRANSFORMER VAULT.
6. INSTALL MH2 AND MH3 AT LOCATIONS SHOWN. USE EXTREME CAUTION WHEN INSTALLING MH3 SO AS NOT TO DAMAGE THE EXISTING PRIMARY CABLE SERVING TRANSFORMER VAULT #04148 TO BLDG 6993.
7. INSTALL PRIMARY DUCTLINE FROM MH1 TO MH2 AND MH3. USE EXTREME CAUTION NOT TO DAMAGE THE PRIMARY CABLES SERVING BLDG 6993. HAND DIG AS NECESSARY.
8. INSTALL PRIMARY CABLES FROM JUNCTION PEDESTAL #A9JP2–6995 TO MH1, THEN FROM THE 4 WAY JUNCTION TO THE NEW TRANSFORMER FROM MADIGAN AND TO TRANSFORMER VAULT #04148 TO BLDG 6993 VIA MH2 AND MH3.
9. INSTALL SECONDARY FEEDER FROM THE NEW TRANSFORMER AT MADIGAN TO THE BUILDING MAIN DISTRIBUTION PANEL "MDP". ALSO INSTALL NEW CABLE TO TRANSFORMER VAULT #04148.
10. REMOVE EXISTING PRIMARY CABLES THAT FED THE EXISTING TRANSFORMER AND TRANSFORMER VAULT #04148.
11. REMOVE EXISTING MH IN THE GRASSY PLAYGROUND AREA. ABANDON EXISTING CONDUITS IN PLACE AND AT THE LOCATION OF THE MH, SEAL WITH CONCRETE AT THE MANHOLE LOCATION.
12. AFTER SWITCH–OVER OF POWER REMOVE EXISTING PAD MOUNTED TRANSFORMER PRIMARY CABLES AND SECONDARY CABLES. ABANDON CONDUIT IN PLACE. NOTIFY JBLM BASE FACILITIES (2) WEEKS IN ADVANCE PRIOR TO POWER SHUT–DOWN. ALL SCHEDULED POWER SHUT–DOWN AND POWER OUTAGES SHALL ONLY BE ACCOMPLISHED DURING SATURDAYS AND SUNDAYS. POWER OUTAGE SHALL BE LIMITED TO A MAXIMUM OUTAGE FOR ONE WEEKEND DURING SATURDAY AND SUNDAY. TEMPORARY ON–SITE STANDBY POWER GENERATOR MAY BE NEEDED, COORDINATE WORK WITH USACE. SUBMIT OUTAGE PLAN PRIOR TO COMMENCEMENT OF WORK FOR APPROVAL.
13. TURN OVER EXISTING TRANSFORMER AND OTHER REMOVED EQUIPMENT TO JBLM BASE FACILITIES DEPARTMENT OR DISPOSE AS DIRECTED.
14. PROVIDE DOUBLE DIAMOND PLATE MANHOLE COVERS FOR MANHOLES WITH JUNCTIONS. UTILITY VAULT 776 MANHOLE W/ NO. 77–2–332P, VERIFY WITH JBLM BEFORE ORDERING TO ENSURE CORRECT MANHOLE. ALL MANHOLE SHALL BE UTILITY VAULT 776 MANHOLE PER JBLM STANDARDS.
15. COORDINATE WITH JBLM ELECTRICAL FOR ALL PRIMARY AND POWER OUTAGES, INSPECTIONS, AND CONSTRUCTION STANDARDS AND REQUIREMENTS PRIOR TO COMMENCEMENT OF WORK.

AS BUILT DRAWINGS

CB ENGINEERS
Building Experience

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Kent, WA. 98032
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Fax: 253.235.2900
www.holmes.com

US Army Corps of Engineers
Seattle District

Symbol	Description	Date	Appr.

Symbol	Description	Appr.

Designed by:	VW	Date:	22 OCT 2012
Drawn by:	VW	Design file no.	
Checked by:	MK	Rev.	
Submitted by:			
Chief,		Section	

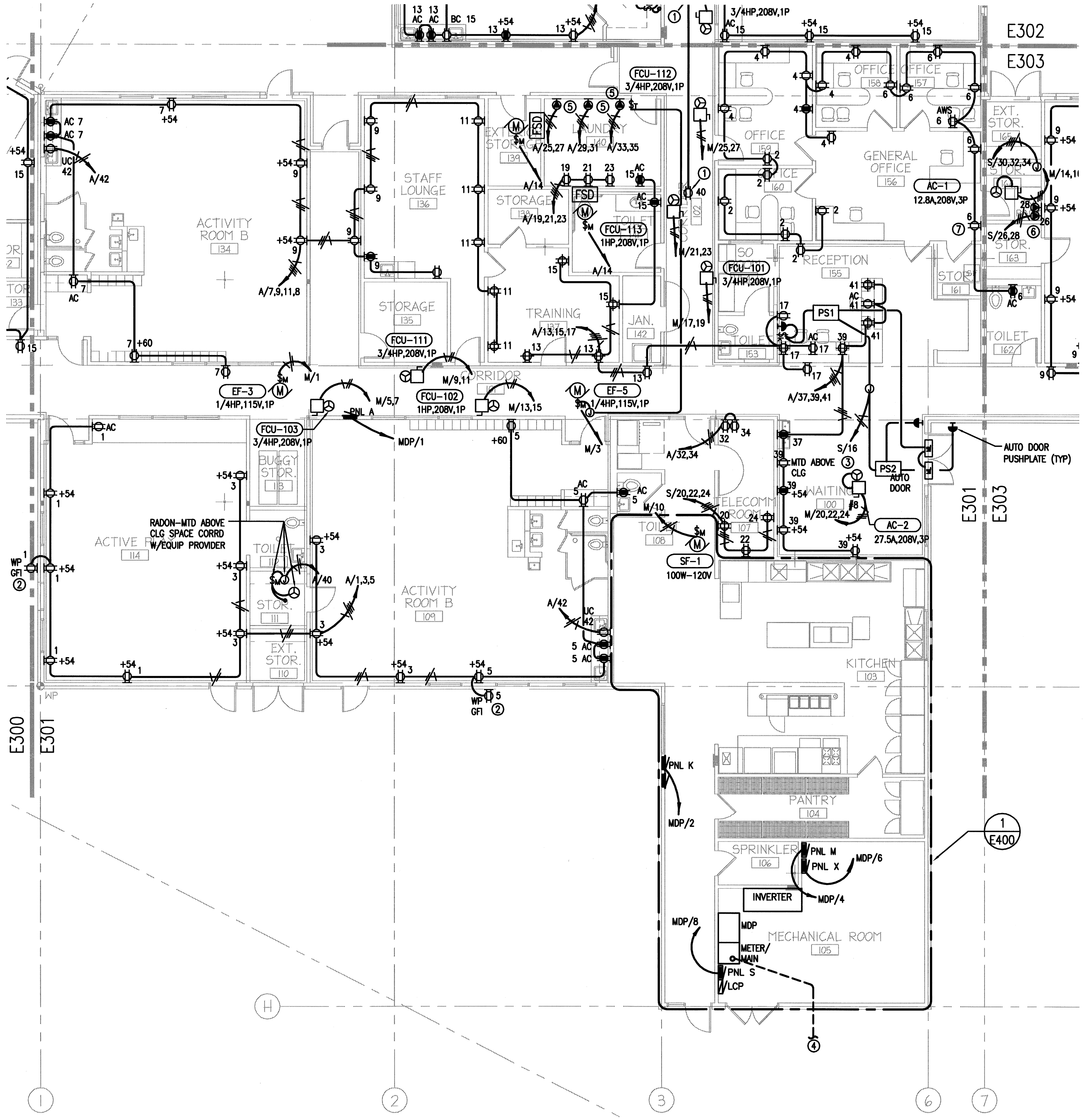
U.S. ARMY ENGINEER DISTRICT, SEATTLE
CORPS OF ENGINEERS
SEATTLE, WASHINGTON
Submitted by:
BergerABAM
(206) 431-2300
FAX:(206) 431-2250
Principal

FY09 ADDITION/RENOVATION
CHILD DEVELOPMENT CENTER
AGES 0-5 YEARS

MADIGAN CDC
ELECTRICAL SITE PLAN

FT. LEWIS PN 070102 WASHINGTON

PLATE NUMBER:
E100



ENLARGED POWER PLAN

SCALE: 1/8"=1'-0"

1/8"=1'-0" 0 2 4 6 8 12 16 24 SCALE FEET

GENERAL NOTES:

1. WHERE DEVICE OUTLETS ARE SHOWN BACK TO BACK IN PARTITION WALLS. OULETS SHALL BE LOCATED A MINIMUM OF 6" APART, OUTSIDE EDGE TO OUTSIDE EDGE.
2. EXTERIOR WEATHERPROOF RECEPTACLES TO BE MOUNTED AT +18" AFF.
3. REPLACE THE GFCI ELECTRICAL OUTLETS LOCATED IN THE REFRIGERATOR RECESSES UNDER THE FOOD PREP COUNTER IN ALL CHILD ACTIVITY ROOMS WITH NON-GFCI RECEPTACLES. GFCI PROTECTION FOR PERSONNEL IS NOT REQUIREDWHEN THE OUTLET IS NOT READILY ACCESSIBLE TO PERSONNEL. THE INDER-COUNTER ELECTRICAL OUTLETS WILL BE PROTECTED BY THE REFRIGERATORS THAT WILL BE LOCATED IN THOSE RECESSES.

SECURITY SYSTEM

1. EVERGREEN FIRE AND SECURITY SHALL REMOVE AND PROVIDE THE INTEGRATED COMMERCIAL INTRUSION DETECTION SSYTEM (ICIDS DURESS BUTTON UNDER THE CUSTOMER SERVICE COUNTER AT THE MAIN ENTRY. REMOVAL OF THE DURESS BUTTON SHALL TAKE PLACE IMMEDIATELY. REINSTALLATION AND PERFORMANCE TESTING OF THE DURESS SYSTEM SHALL TAKE PLACE UPON COMPLETION OF INSTALLATION OF NEW COUNTER. PROVIDE CONDUIT AND WIRING FOR THE DEDICATED CIRCUIT AND PROVIDE CONDUIT AND WIRING FROM THE RTU PANEL TO THE EMERGENCY BUTTON. LOW VOLTAGE WIRING SHALL BE COMPLETED BY EVERGREEN FIRE AND SECURITY AND THE 120-VOLT WIRING SHALL BE COMPLETED BY THE ELECTRICAL SUBCONTRACTOR

SHEET NOTES:

(1) MOUNT RECEPTACLE AT FCU PLATFORM. COORDINATE LOCATION AND REQUIREMENTS WITH MECHANICAL PRIOR TO ROUGH-IN.

(2) REMOVE, REPLACE EXISTING DUPLEX RECEPTACLE AND RECONNECT AS SHOWN.

(3) DOOR MANAGEMENT SYSTEMS, SEE DETAIL ON THIS SHEET.

(4) (2) 4"CONDUIT TO NEW PAD MOUNTED TRANSFORMER. SEE SITE PLAN ON SHEET E100. COORDINATE WITH STRUCTURAL FOR INSTALLATION OF CONDUITS IN CONCRETE FLOOR SLAB.

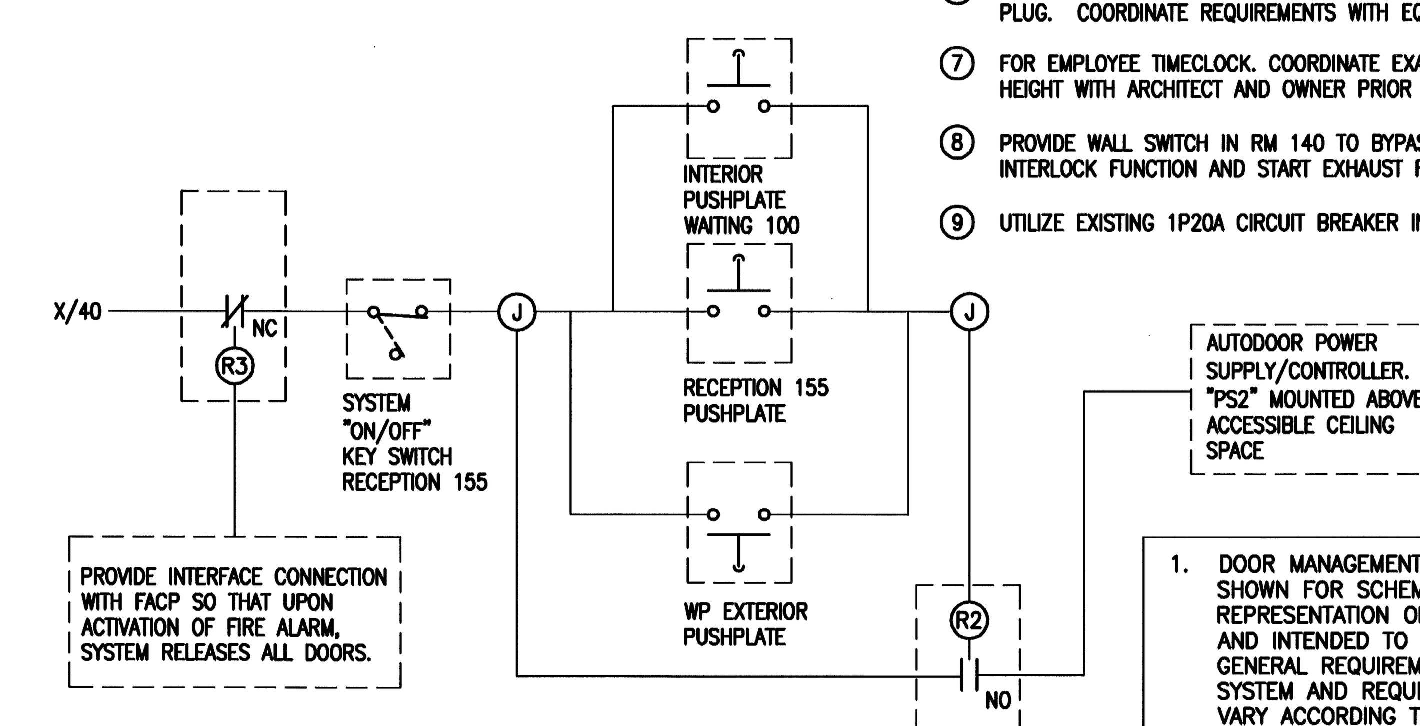
(5) REMOVE AND REPLACE EXISTING DRYER WITH NEW NEMA 6-30R. CONNECT TO NEW CIRCUIT SHOWN. PROVIDE MATCHING POWER CORD AND PLUG. COORDINATE REQUIREMENTS WITH EQUIPMENT VENDOR.

(6) NEMA L5-30P3W125V@ 18"AFF. PROVIDE MATCHING POWER CORD AND PLUG. COORDINATE REQUIREMENTS WITH EQUIPMENT VENDOR.

(7) FOR EMPLOYEE TIMECLOCK. COORDINATE EXACT LOCATION AND MOUNTING HEIGHT WITH ARCHITECT AND OWNER PRIOR TO ROUGH-IN.

(8) PROVIDE WALL SWITCH IN RM 140 TO BYPASS THE DRYER/FAN INTERLOCK FUNCTION AND START EXHAUST FAN EF-5.

(9) UTILIZE EXISTING 1P20A CIRCUIT BREAKER IN POSITION 40 IN PANEL "A".

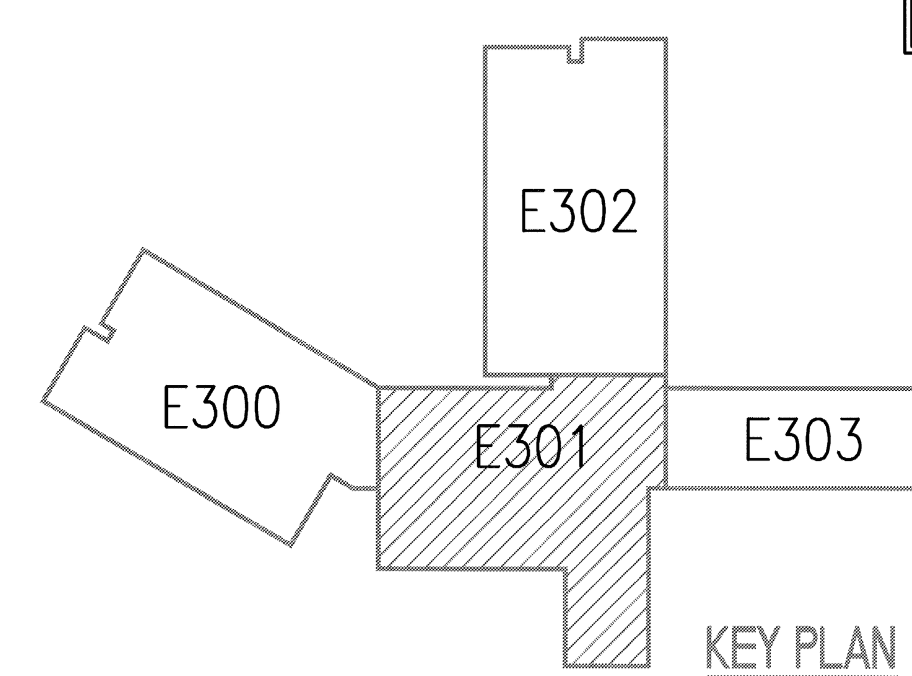


1. DOOR MANAGEMENT DETAIL ARE SHOWN FOR SCHEMATIC REPRESENTATION OF THE SYSTEM AND INTENDED TO SHOW ONLY THE GENERAL REQUIREMENTS. ACTUAL SYSTEM AND REQUIREMENTS MAY VARY ACCORDING TO APPROVED DOOR VENDOR/MANUFACTURER.
2. CONTRACTOR TO PROVIDE ELECTRIC LATCH, ELECTRIC HINGE AND ALL OTHER ITEMS REQUIRED TO PROVIDE EMERGENCY PUSHBAR EXIT OVERRIDE.

2 E301 DOOR MANAGEMENT DETAIL

SCALE: NOT TO SCALE

AS BUILT DRAWINGS



KEY PLAN

CB ENGINEERS Building Experience

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Kent, WA. 98032
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Fax: 253.235.2900
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US Army Corps of Engineers
Seattle District

Symbol	Description	Date	Appr.

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Designed by: VW	Date: 22 OCT 2012	Drawn by: VW	Design file no.
Checked by: MK	Rev.	Submitted by:	Chief, ______ Section

U.S. ARMY ENGINEER DISTRICT, SEATTLE
CORPS OF ENGINEERS
SEATTLE, WASHINGTON
Submitted by: BergerABAM
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Principal

FY09 ADDITION/RENOVATION
CHILD DEVELOPMENT CENTER
AGES 0-5 YEARS
MADIGAN CDC
ENLARGED POWER PLAN
FT. LEWIS PN 070102 WASHINGTON

PLATE NUMBER:
E301

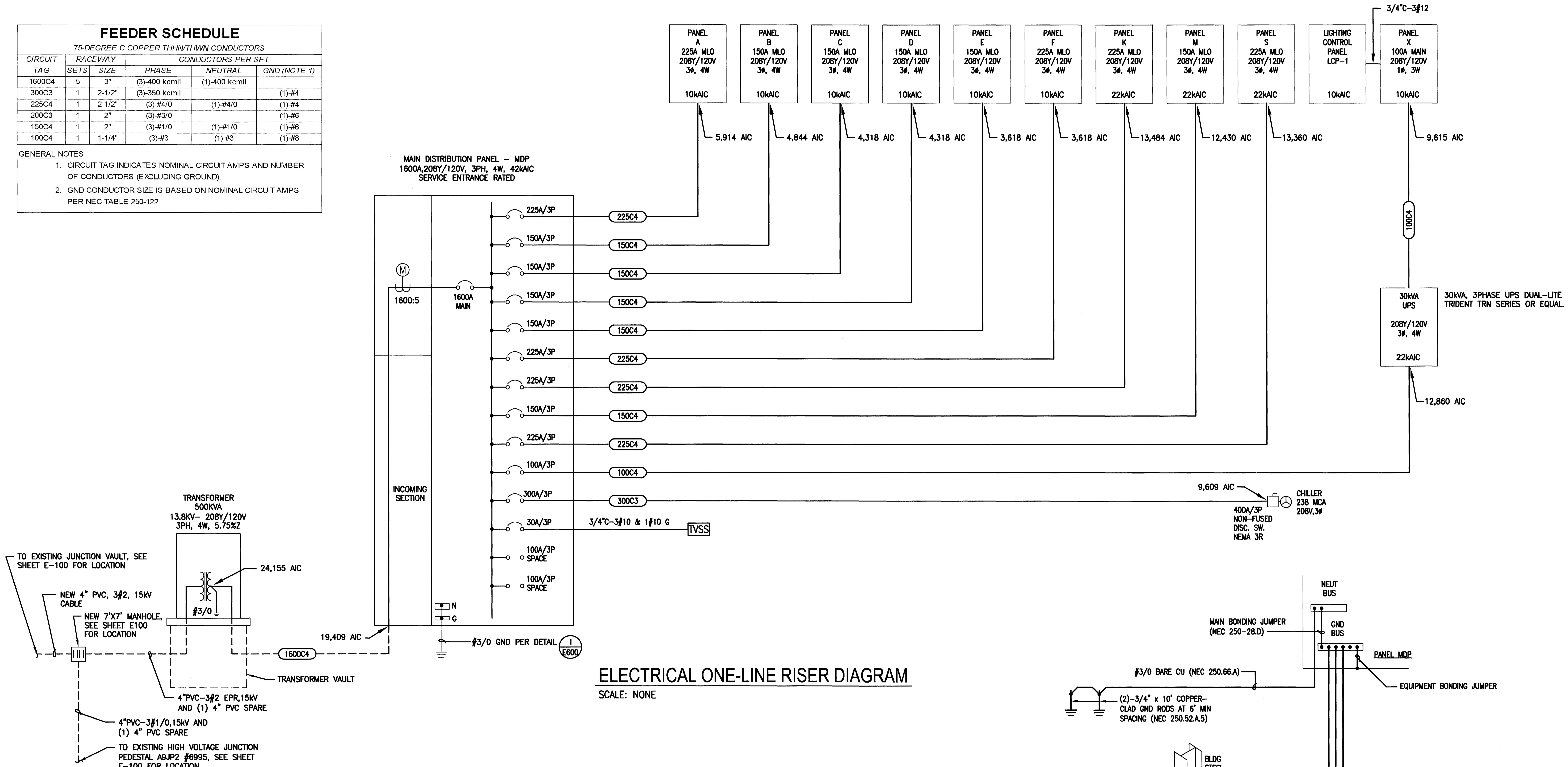
FEEDER SCHEDULE

75-DEGREE C COPPER THHN/THWN CONDUCTORS

CIRCUIT TAG	RACEWAY		CONDUCTORS PER SET		
	SETS	SIZE	PHASE	NEUTRAL	GND (NOTE 1)
1600C4	5	3"	(3)-400 kcmil	(1)-400 kcmil	
300C3	1	2-1/2"	(3)-350 kcmil		(1)-#4
225C4	1	2-1/2"	(3)-#4/0	(1)-#4/0	(1)-#4
200C3	1	2"	(3)-#3/0		(1)-#6
150C4	1	2"	(3)-#1/0	(1)-#1/0	(1)-#6
100C4	1	1-1/4"	(3)-#3	(1)-#3	(1)-#8

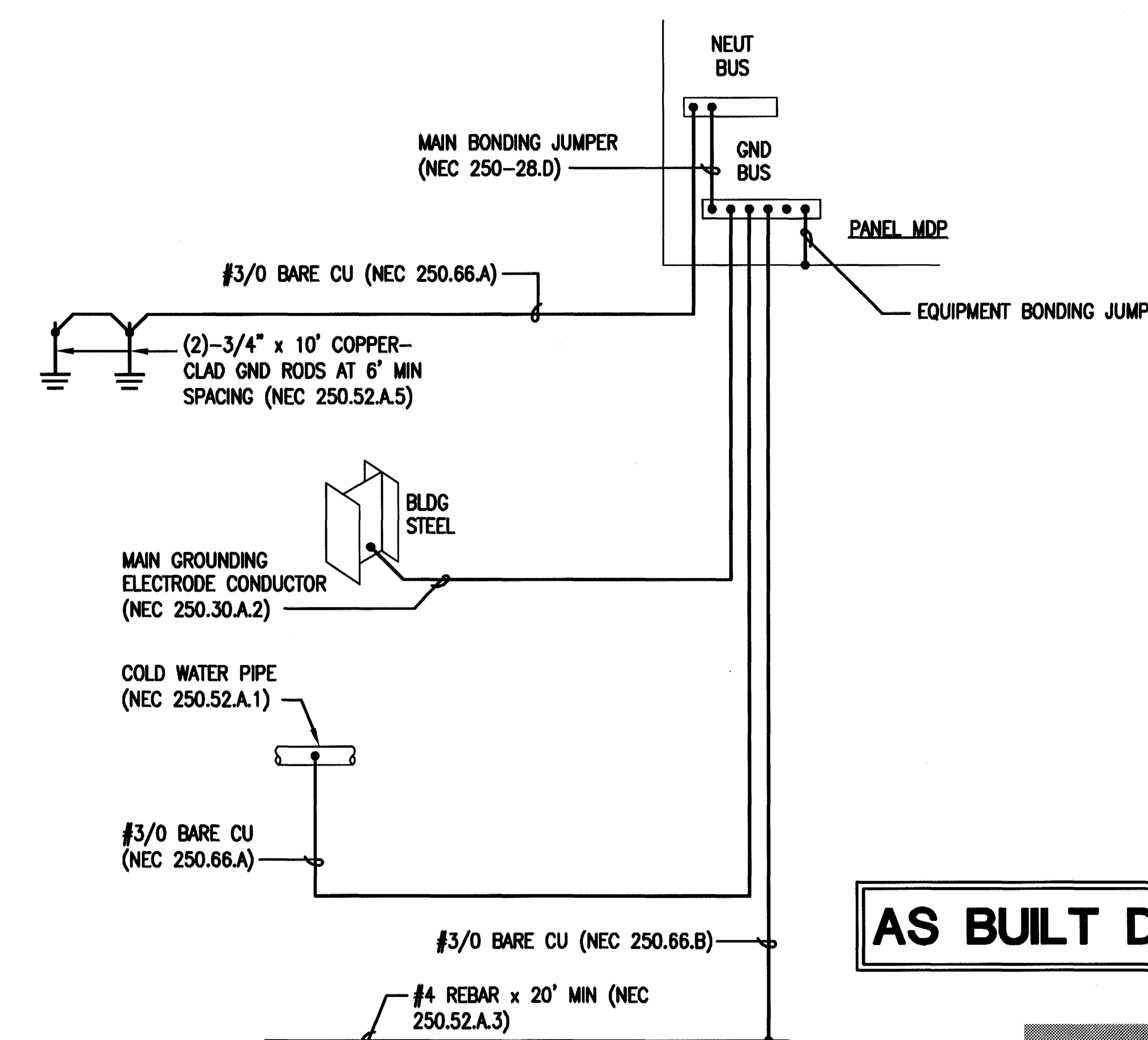
GENERAL NOTES

1. CIRCUIT TAG INDICATES NOMINAL CIRCUIT AMPS AND NUMBER OF CONDUCTORS (EXCLUDING GROUND).
2. GND CONDUCTOR SIZE IS BASED ON NOMINAL CIRCUIT AMPS PER NEC TABLE 250-122



ELECTRICAL ONE-LINE RISER DIAGRAM

SCALE: NONE



NOTE: AT A MINIMUM FOR GROUNDING SYSTEM, PROVIDE GROUND ELECTRODE FOR CONNECTION TO GROUND BUS AND COLD WATER METALLIC PIPE IF BUILDING STEEL AND UFER GROUND TERMINATION IS NOT PRACTICABLE.

1 E600 GROUNDING DIAGRAM

SCALE: NONE

AS BUILT DRAWINGS

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FY09 ADDITION/RENOVATION
CHILD DEVELOPMENT CENTER
AGES 0-5 YEARS

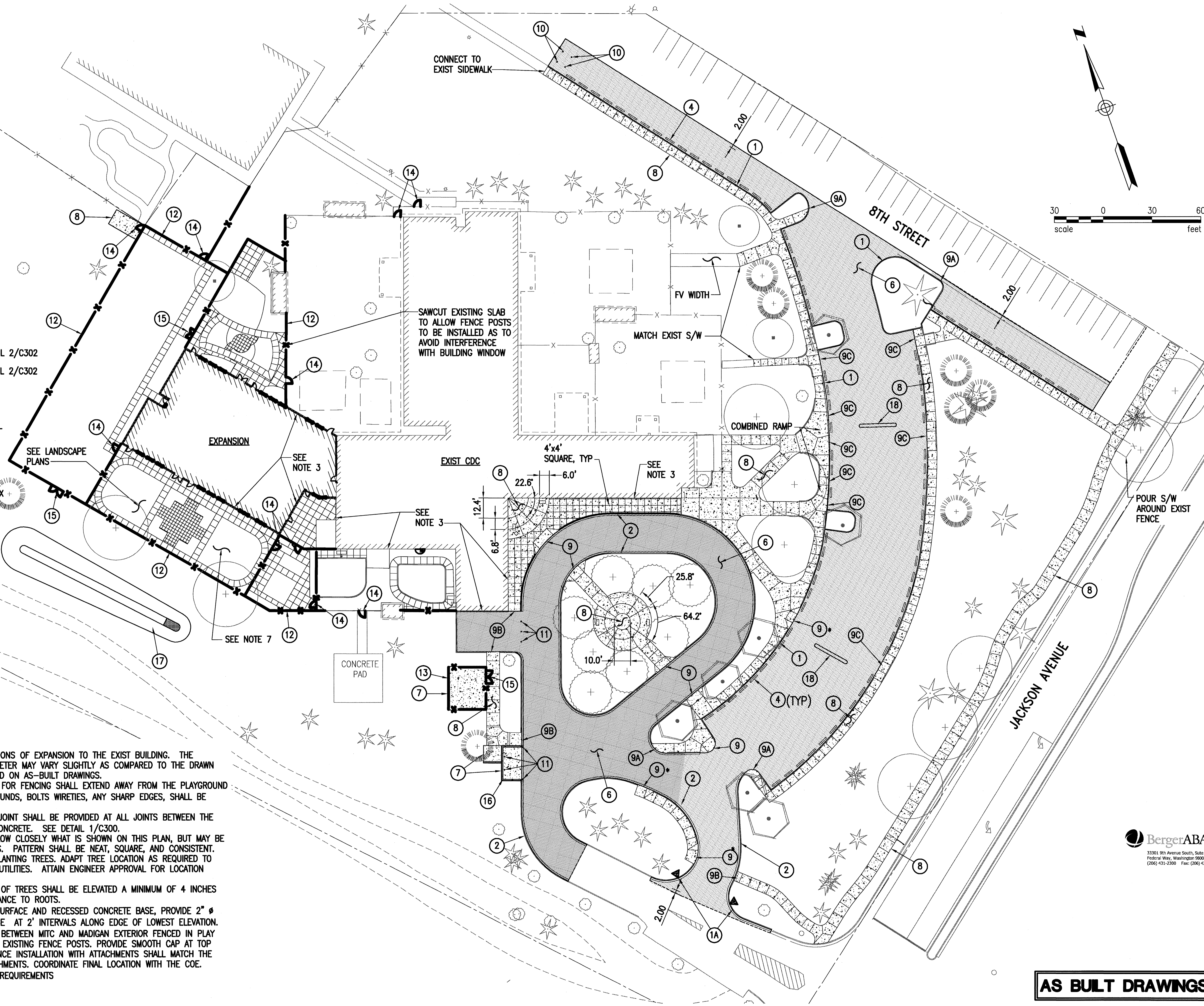
MADIGAN CDC
ONE-LINE RISER DIAGRAM

FT. LEWIS PN 070102 WASHINGTON

PLATE NUMBER:
E600

1. CONC CURB ONLY – DETAIL 5/C300
1A. CURB CUT – DETAIL 7/C300, SYMBOL ▼
2. CONC CURB AND GUTTER – DETAIL 4/C300
3. CONC GUTTER – DETAIL 9/C300 (NOT USED)
4. CONC CURB BUMPER – DETAIL 6/C300
5. EXTRUDED CURB – DETAIL 3/C301 (NOT USED)
6. AC PAVEMENT – DETAIL 8/C300
7. REINFORCED CONCRETE – DETAIL 1/C301
8. SIDEWALK – DETAIL 2/C301, 2/C300
9. SIDEWALK RAMP – DETAIL 8/C301
9A. LIMITED RAMP – DETAIL 6/C301
9B. PERPENDICULAR RAMP – DETAIL 5/C301
9C. PARALLEL RAMP – DETAIL 7/C301
10. STANDARD BOLLARD – DETAIL 4/C303
11. REMOVABLE BOLLARD – DETAIL 4/C302
12. 4' BLACK VINYL CHAIN LINK FENCE – DETAIL 2/C302
13. 6' BLACK VINYL CHAIN LINK FENCE – DETAIL 2/C302
14. SINGLE GATE – DETAIL 1/C302
15. DOUBLE GATE – DETAIL 3/C302
16. CMU ENCLOSURE – SEE STRUCTURAL DETAIL
17. BIO-FILTRATION SWALE – DETAIL 2/C307
18. SPEED BUMP – DETAIL 9/C301

* SYMBOL INDICATES WINGS OF RAMP SHALL BE 1:12 SLOPE MAX



NOTES:

1. FIELD VERIFY AND FIELD FIT EXACT DIMENSIONS OF EXPANSION TO THE EXIST BUILDING. THE SURVEYED LOCATION OF THE BUILDING PERIMETER MAY VARY SLIGHTLY AS COMPARED TO THE DRAWN ARCHITECTURAL FLOOR PLAN, WHICH IS BASED ON AS-BUILT DRAWINGS.
2. THREADED ENDS OF BOLTS AND WIRE TIES FOR FENCING SHALL EXTEND AWAY FROM THE PLAYGROUND (OUTSIDE). FOR FENCING BETWEEN PLAYGROUNDS, BOLTS WIRETIES, ANY SHARP EDGES, SHALL BE GROUND SMOOTH OR CAPPED OR CAULKED.
3. A THICKENED EDGE CONCRETE EXPANSION JOINT SHALL BE PROVIDED AT ALL JOINTS BETWEEN THE BUILDING FOUNDATION AND THE ADJOINING CONCRETE. SEE DETAIL 1/C300.
4. PATTERN OF CONCRETE JOINTS SHALL FOLLOW CLOSELY WHAT IS SHOWN ON THIS PLAN, BUT MAY BE ADAPTED BASED ON FIELD RECOMMENDATIONS. PATTERN SHALL BE NEAT, SQUARE, AND CONSISTENT.
5. FIELD VERIFY UTILITY LOCATIONS BEFORE PLANTING TREES. ADAPT TREE LOCATION AS REQUIRED TO MAINTAIN 10-FT CLEARANCE FROM EXISTING UTILITIES. ATTAIN ENGINEER APPROVAL FOR LOCATION ADJUSTMENTS OR DELETION.
6. SIDEWALKS INSTALLED WITHIN ROOT ZONES OF TREES SHALL BE ELEVATED A MINIMUM OF 4 INCHES ABOVE EXISTING GRADE TO REDUCE DISTURBANCE TO ROOTS.
7. FOR ALL PLAYGROUND AREAS WITH FOAM SURFACE AND RECESSED CONCRETE BASE, PROVIDE 2" ø WEEP HOLES TO BASE COURSE FOR DRAINAGE AT 2' INTERVALS ALONG EDGE OF LOWEST ELEVATION.
8. PROVIDE A NEW 52'-0"L x 4'-0"H FENCE BETWEEN MITC AND MADIGAN EXTERIOR FENCED IN PLAY AREAS. THE NEW FENCE SHALL CONNECT TO EXISTING FENCE POSTS. PROVIDE SMOOTH CAP AT TOP AND BOTTOM OF NEW FENCE POST. NEW FENCE INSTALLATION WITH ATTACHMENTS SHALL MATCH THE ADJACENT TERMINAL FENCE POST AND ATTACHMENTS. COORDINATE FINAL LOCATION WITH THE COE. REFERENCE DETAIL 2/C302 FOR ADDITIONAL REQUIREMENTS

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AS BUILT DRAWINGS

US Army Corps of Engineers
Seattle District

Designed by: DVJ	Date: 22 OCT 2012
Drawn by: DLY	Design file no.
Checked by: SEE	Rev.
Submitted by:	
Chief,	Section

U.S. ARMY ENGINEER DISTRICT, SEATTLE
CORPS OF ENGINEERS
SEATTLE, WASHINGTON

Submitted by: BergerABAM (206) 431-2300 FAX (206) 431-2250

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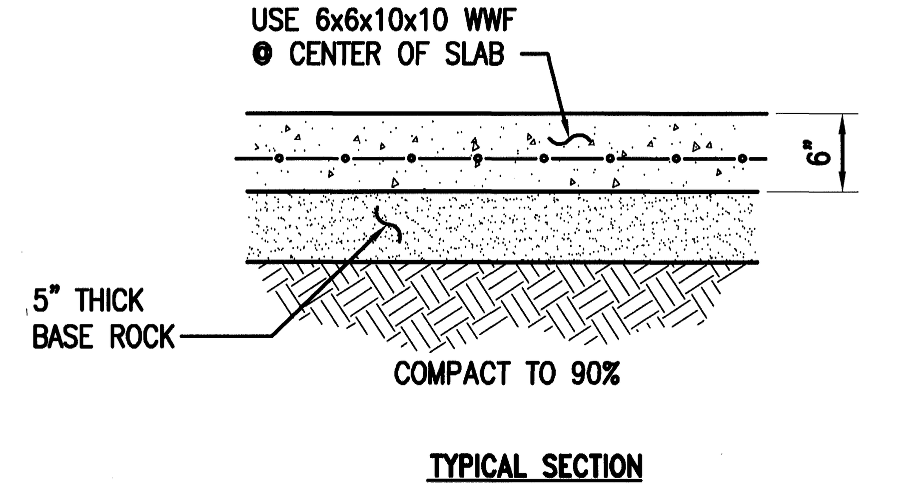
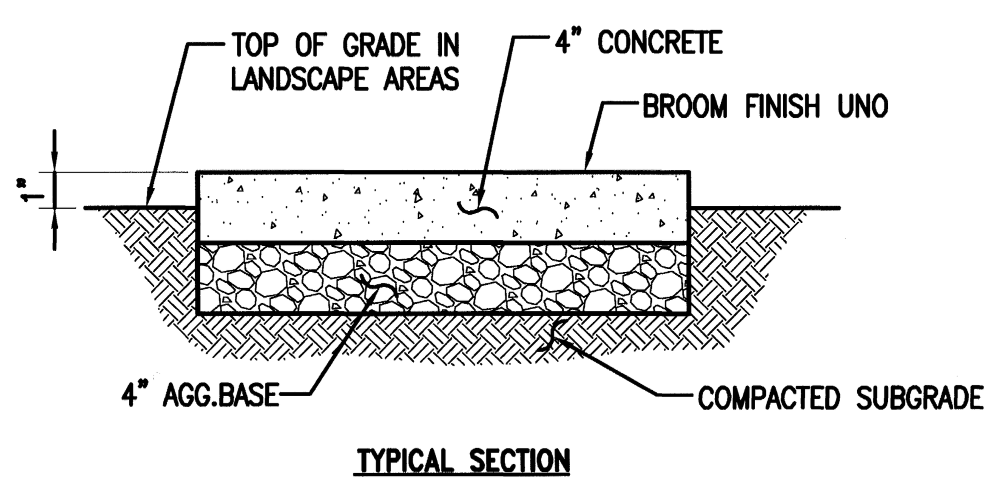
FY09 ADDITION/RENOVATION
CHILD DEVELOPMENT CENTER
AGES 0-5 YEARS
MADIGAN CDC
SITE PLAN
FT. LEWIS PN 070102 WASHINGTON

PLATE NUMBER:
C103

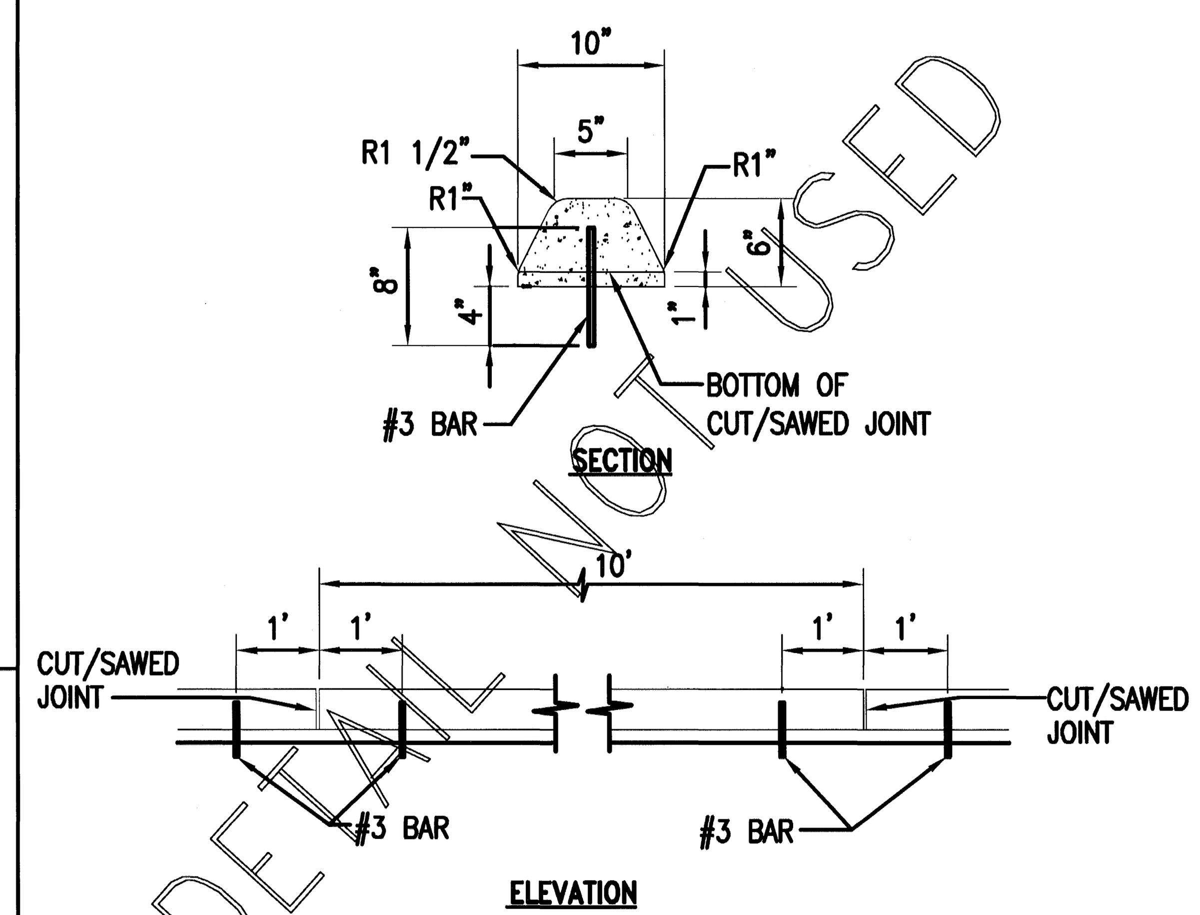
Sheet 10 of 212



1 / C301 CONC. PAD/EMERGENCY VEHICLE ACCESS
SCALE: N.T.S.



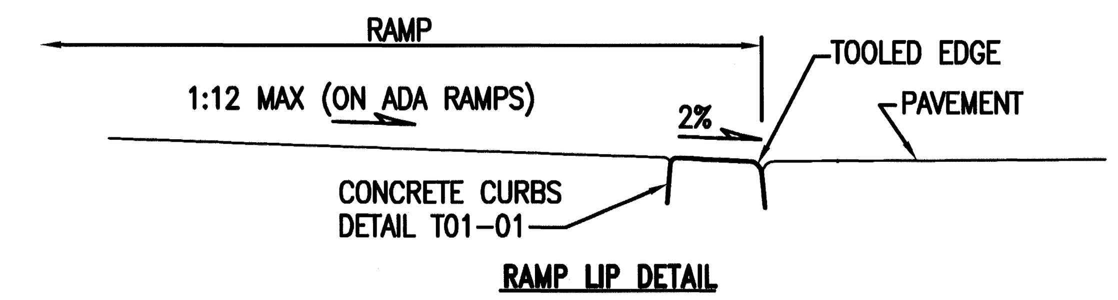
2 / C301 SIDEWALK
SCALE: N.T.S.



NOTES:

1. THE NON-SHRINK GROUT USED TO ATTACH THE PRECAST CURB SHALL CONSIST OF 3:1 SAND AND CEMENT. GROUT TO A DEPTH OF 1" FROM THE TOP OF THE BUMPER.
2. LOCATIONS FOR WEEPHOLES, AND V-SLOTS SHALL BE IDENTIFIED ON THE CONTRACT DRAWINGS.
3. CONCRETE SHALL HAVE A COMPRESSIVE STRENGTH OF 3000 PSI AT 28 DAYS.

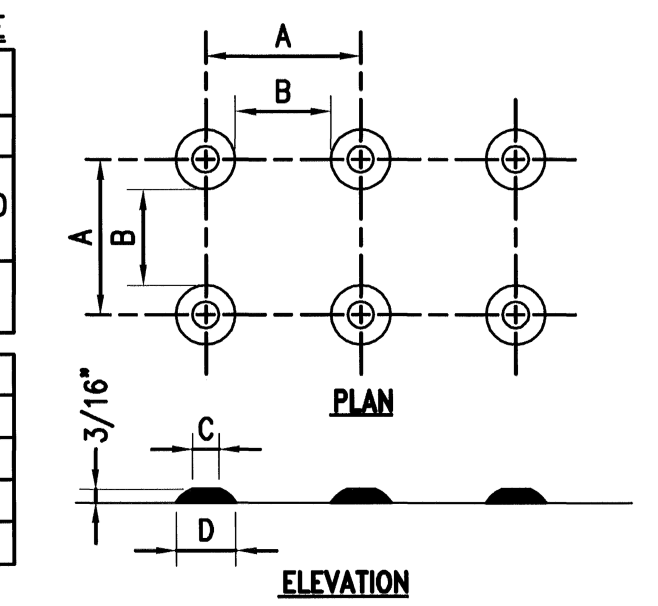
3 / C301 EXTRUDED CURB DETAIL
SCALE: N.T.S.



TRUNCATED DOMES COLOR SCHEME

LOCATION	COLORS
ALL BRICK RAMPS	WHITE
STREETS WITH A MAJORITY OF RESIDENTIAL FRONTAGE	BRICK RED
NON-RESIDENTIAL ARTERIAL STREETS	SAFETY YELLOW

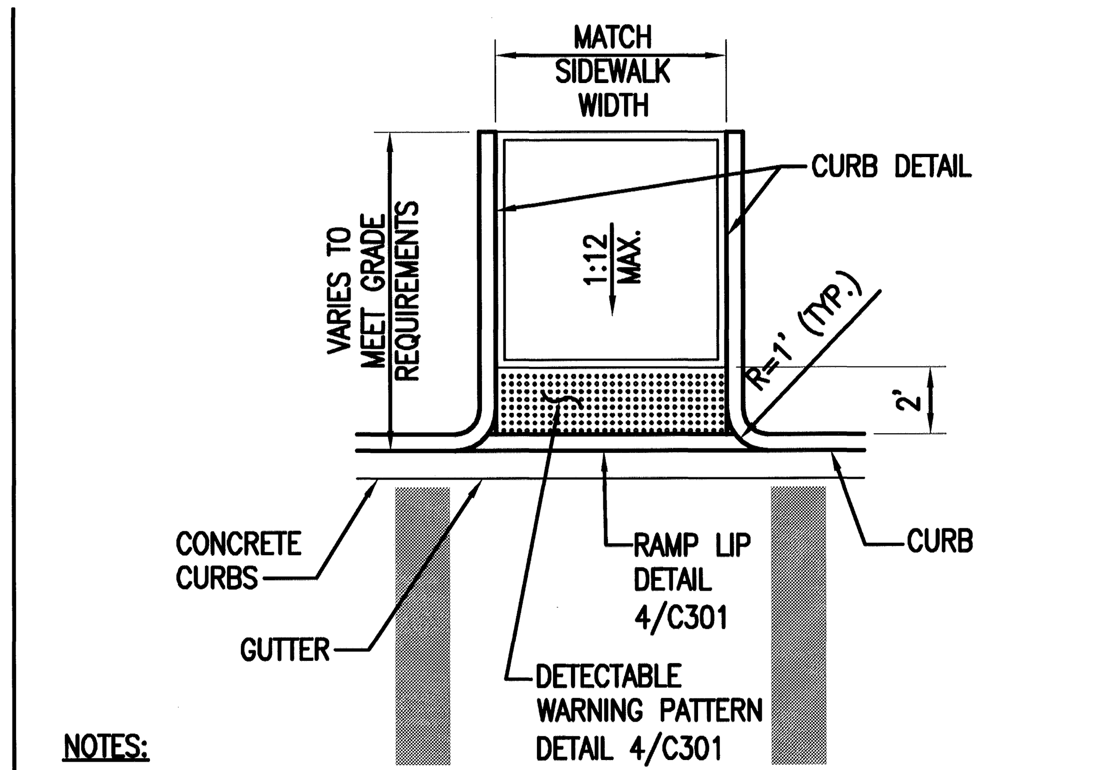
	MIN.	MAX.
A	1 5/8"	2 3/8"
B	2/3"	1 1/2"
C	7/16"	3/4"
D	7/8"	1 7/16"



NOTES:

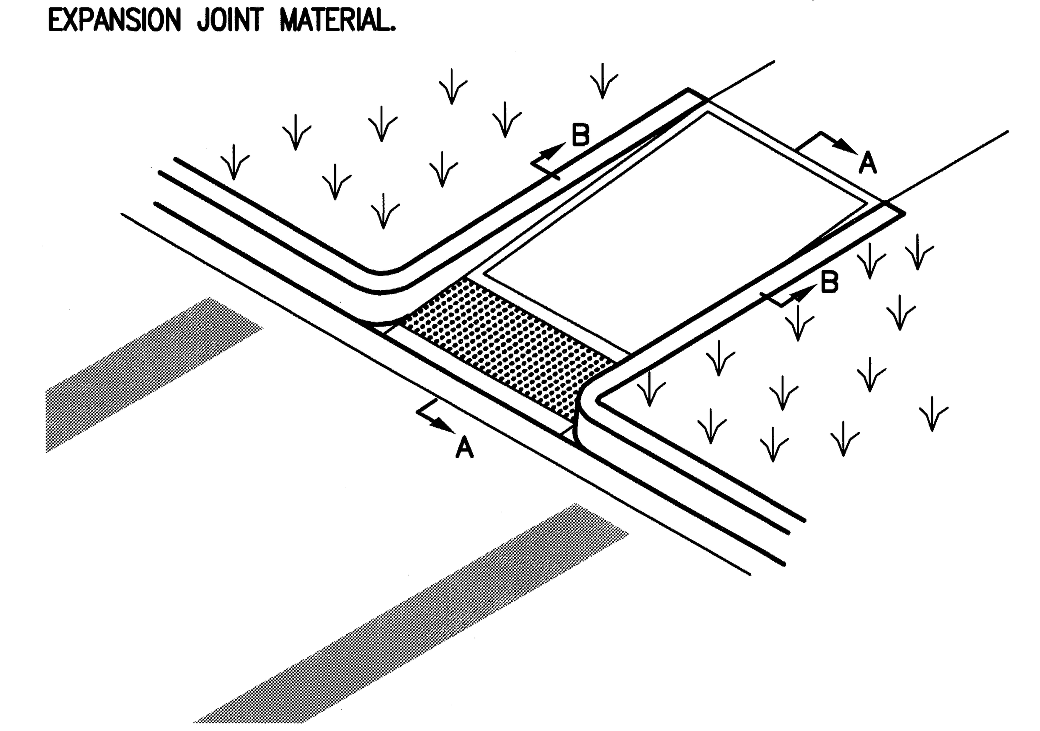
1. MANUFACTURERS SHALL MEET THE REQUIREMENTS LISTED UNDER THE CONTRACT SPECIAL PROVISIONS AND MUST BE ON THE CITY OF VANCOUVER'S APPROVED PRODUCT LIST.
2. DETECTABLE WARNINGS SHALL BE MANUFACTURED USING THE MATERIALS SPECIFIED ON THE PLAN SHEETS WITH THE DOME DIMENSIONS AND SPACING SHOWN AND INSTALLED PER THE MANUFACTURER'S RECOMMENDED PROCEDURES.
3. DETECTABLE WARNINGS SHALL BE INSET INTO NEW CONCRETE. GLUED ON OR NAILED DOWN PRODUCTS ARE NOT ACCEPTABLE FOR NEW CONSTRUCTION.

4 / C301 RAMP LIP AND DETECTABLE WARNING PATTERN DETAIL
SCALE: N.T.S.

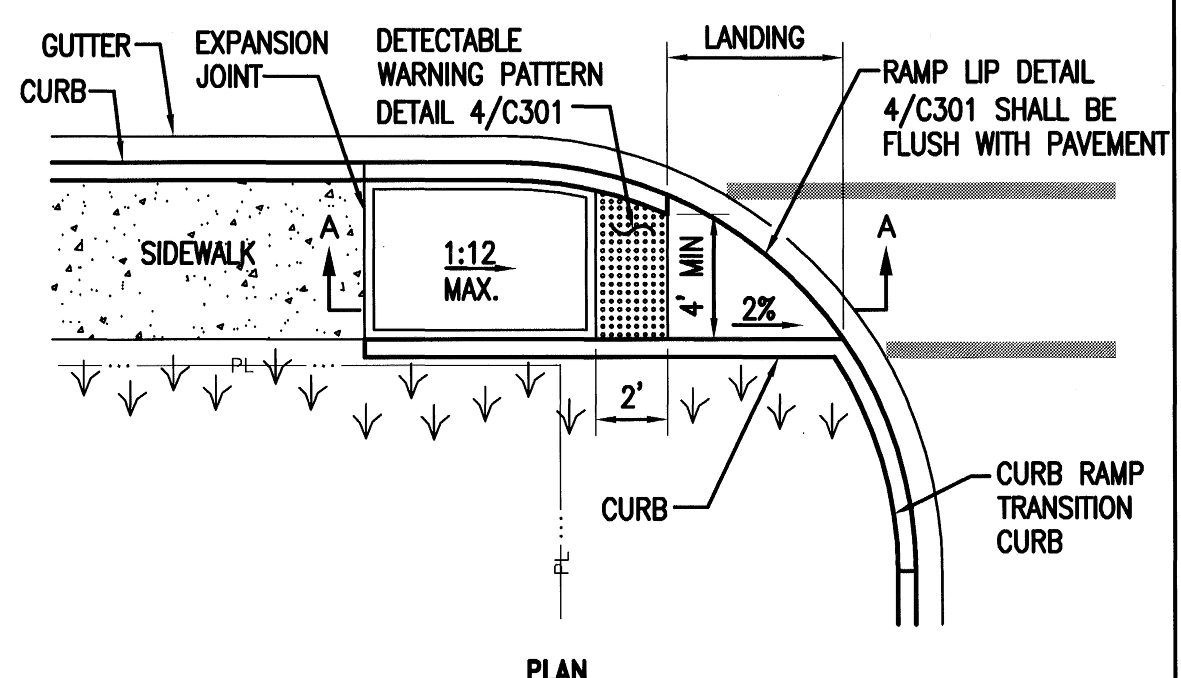


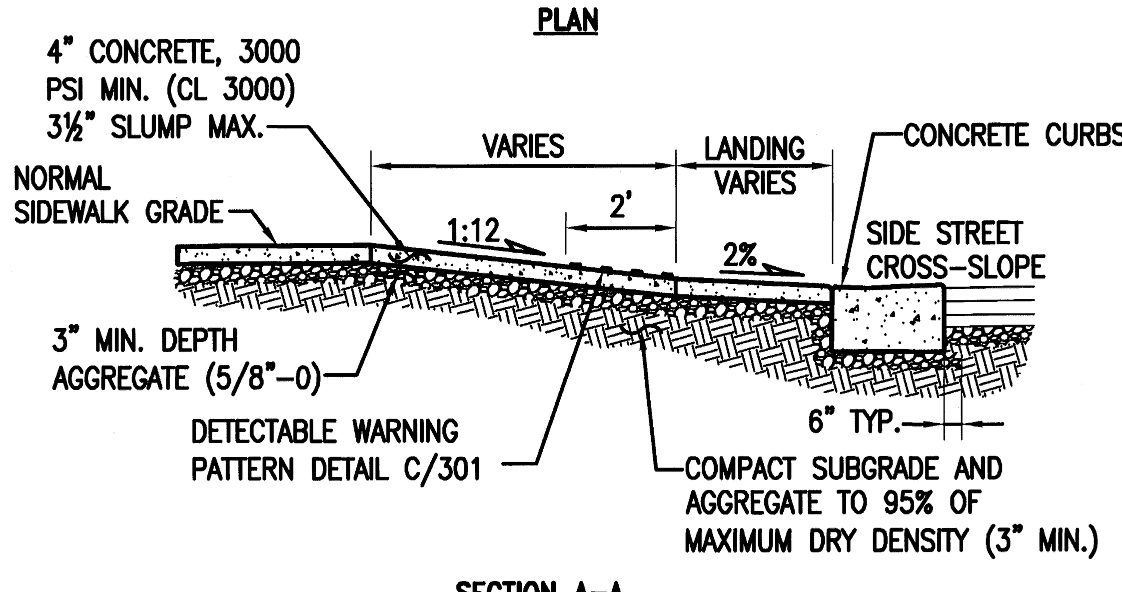
NOTES:

1. RAMP TO BE CENTERED IN CROSSWALK.
2. RAMPS TO BE CONSTRUCTED SEPARATELY FROM SIDEWALK, AND ISOLATED BY EXPANSION JOINT MATERIAL.



5 / C301 PERPENDICULAR RAMP CONSTRUCTION
SCALE: N.T.S.

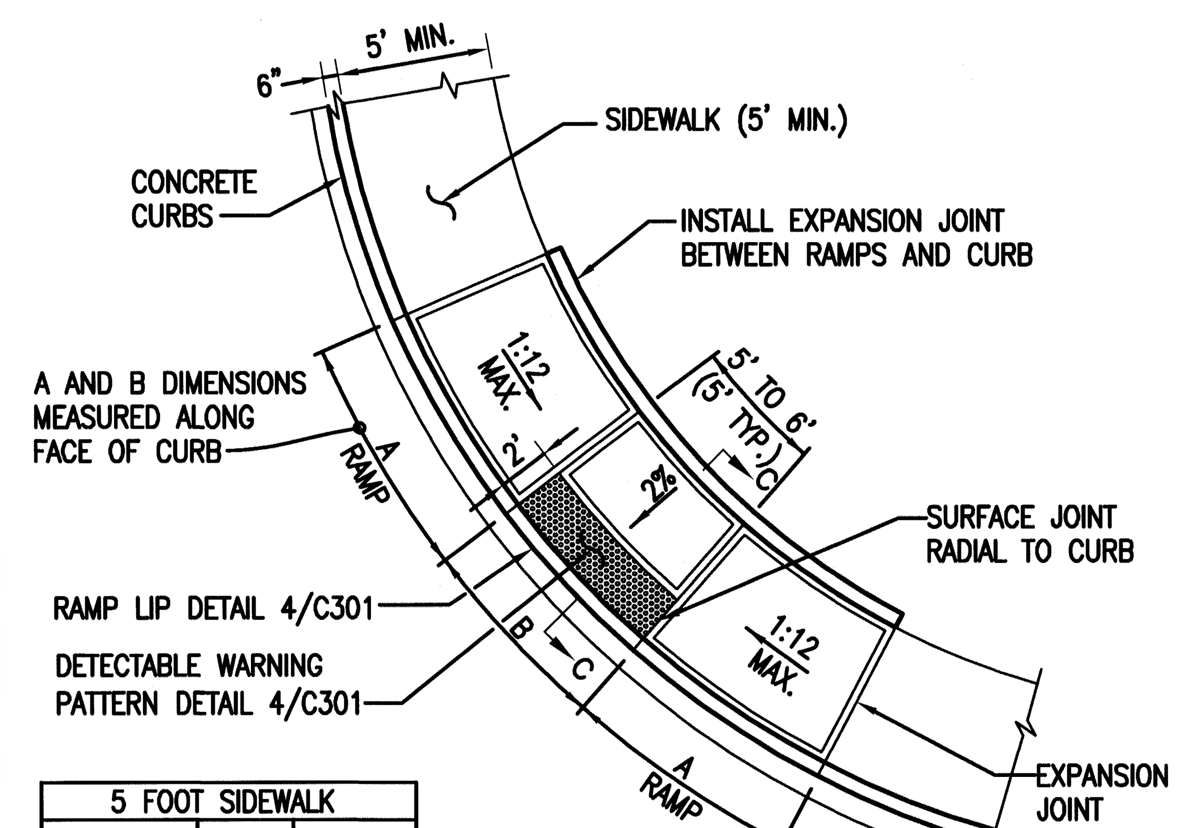




NOTES:

1. RAMP TO BE CENTERED IN CROSSWALK.
2. AN UNOBSTRUCTED PATH OF TRAVEL WITH A MINIMUM WIDTH OF 4' SHALL BE MAINTAINED.
3. IF THE AREA BEHIND THE SIDEWALK IS VEGETATED THE BACK CURB MAY BE REPLACED WITH A SLOPE NO STEEPER THAN 4:1.

6 / C301 CURB RAMP FOR LIMITED R.O.W. AREAS
SCALE: N.T.S.



5 FOOT SIDEWALK		
RADIUS (FT)	A (FT)	B (FT)**
10 *	5.95	5.00
15 *	7.39	5.00
20 *	6.45	5.00
25	6.00	6.41
30	6.00	6.12
35	6.00	5.93
40	6.00	5.80
45	6.00	5.63
∞	6.00	5.00

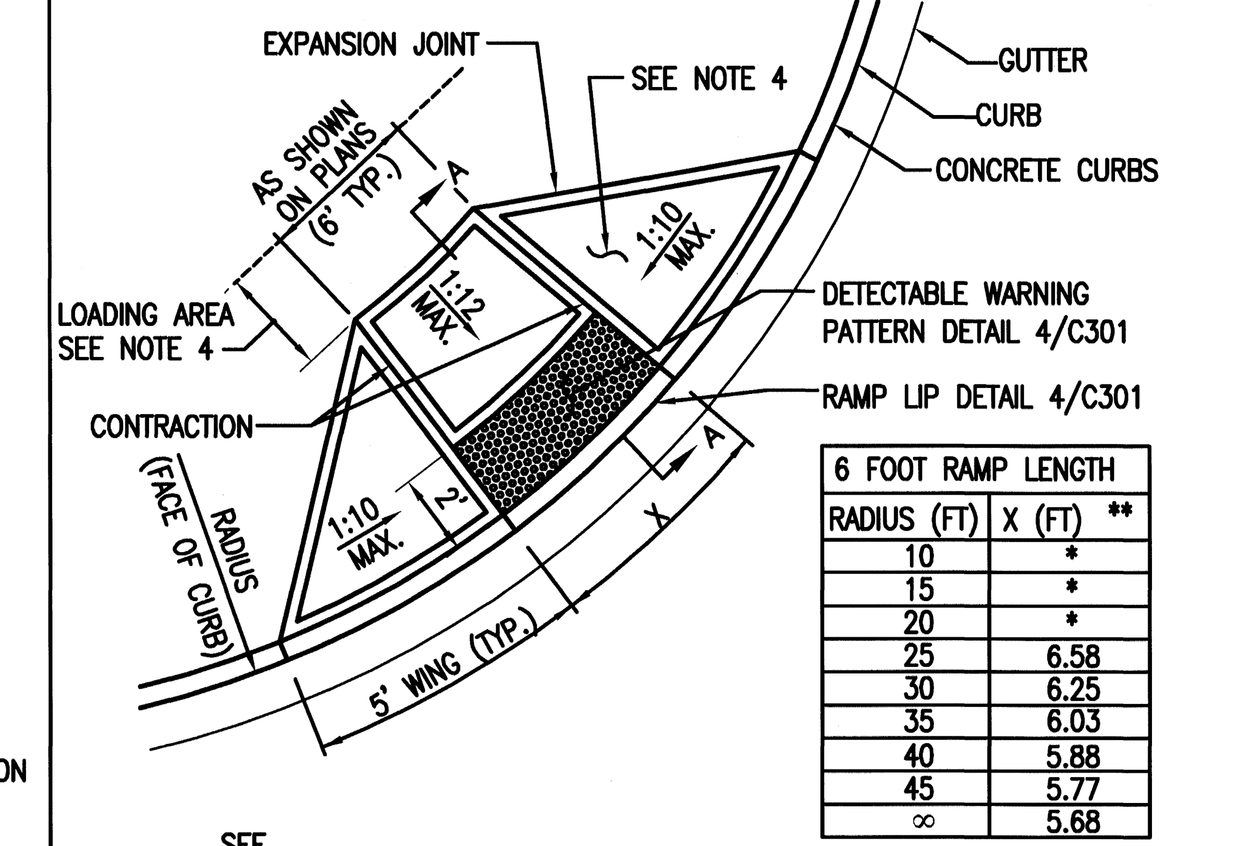
6 FOOT SIDEWALK		
RADIUS (FT)	A (FT)	B (FT)**
10 *	4.90	6.00
15 *	7.63	6.00
20 *	6.58	6.00
25	6.00	6.76
30	6.00	6.38
35	6.00	6.14
40	6.00	5.97
45	6.00	5.84
∞	6.00	5.00

NOTES:

1. RAMPS SHALL HAVE A MAXIMUM 1:12 SLOPE.
2. RAMP TO BE CENTERED IN CROSSWALK.
3. RAMPS TO BE CONSTRUCTED SEPARATELY FROM SIDEWALK AND ISOLATED BY EXPANSION JOINT MATERIAL.
4. IF THE AREA BEHIND THE SIDEWALK IS VEGETATED, THE BACK CURB MAY BE REPLACED WITH A SLOPE NO STEEPER THAN 4:1.

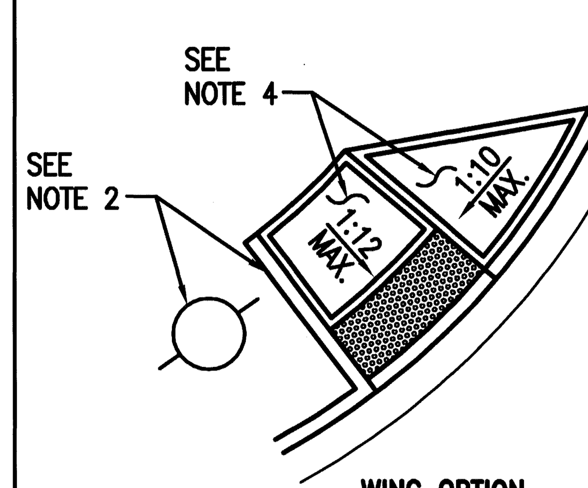
* MAINTAIN SAME WIDTH (5'-6') THRU 1:12 AREA.
** ASSUMED 5' TOP OF RAMP WIDTH.

7 / C301 PARALLEL RAMP CONSTRUCTION
SCALE: N.T.S.



6 FOOT RAMP LENGTH	
RADIUS (FT)	X (FT) **
10	*
15	*
20	*
25	6.58
30	6.25
35	6.03
40	5.88
45	5.77
∞	5.68

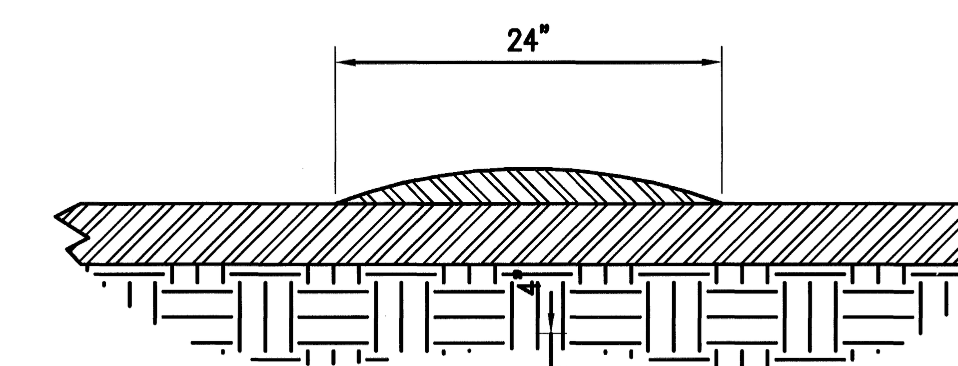
* MAINTAIN SAME WIDTH THRU 1:12 AREA.
** ASSUMED 5' TOP OF RAMP WIDTH.



NOTES:

1. RAMP TO BE CONSTRUCTED SEPARATELY FROM SIDEWALK AND ISOLATED BY EXPANSION JOINT MATERIAL.
2. RAMP WINGS MAY BE REPLACED WITH CURB IF OBSTRUCTION OR PLANTER PREVENTS PEDESTRIAN TRAFFIC IN WING AREA.
3. WING DIMENSIONS MAY VARY TO MEET REQUIRED SLOPE.
4. IF THE LANDING AREA IS LESS THAN 4' IN LENGTH, MAX SLOPE FOR FLARED SIDES OF RAMP IS 1:12.

8 / C301 DIAGNAL RAMP CONSTRUCTION
SCALE: N.T.S.



9 / C301 SPEED BUMP
SCALE: N.T.S.

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AS BUILT DRAWINGS

US Army Corps of Engineers
Seattle District

Symbol	Description	Date	Appr.

Symbol	Description	Date	Appr.

Designed by: DVJ | Date: 22 OCT 2012
Drawn by: DLY | Design file no.
Checked by: SEE | Rev.
Submitted by: | Chief, | Section

U.S. ARMY ENGINEER DISTRICT, SEATTLE
CORPS OF ENGINEERS
SEATTLE, WASHINGTON

Submitted by: BergerABAM (206) 431-2300 FAX (206) 431-2250 Principal

FY09 ADDITION/RENOVATION
CHILD DEVELOPMENT CENTER
AGES 0-5 YEARS
MADIGAN CDC
DETAILS
FT. LEWIS PN 070102 WASHINGTON

PLATE NUMBER:
C301

Sheet 15 of 212

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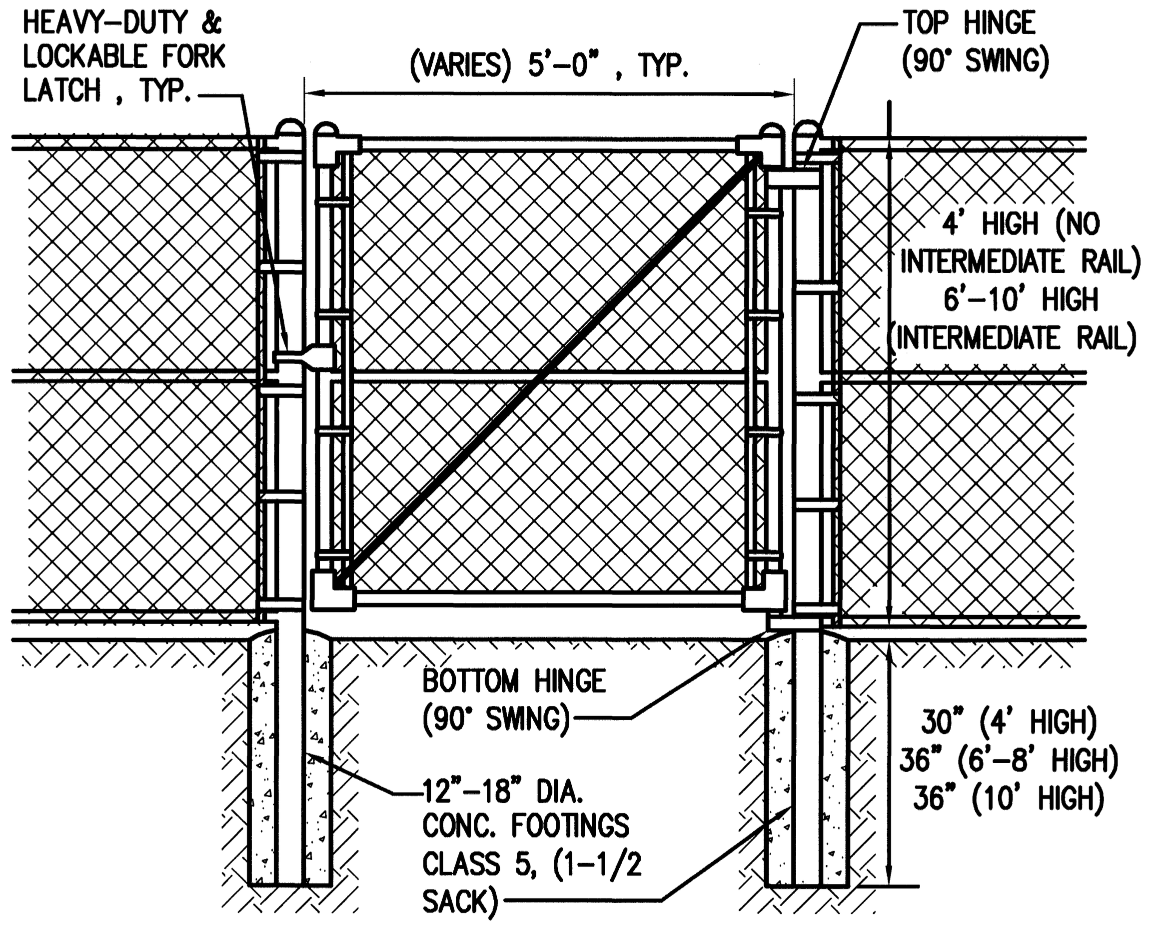
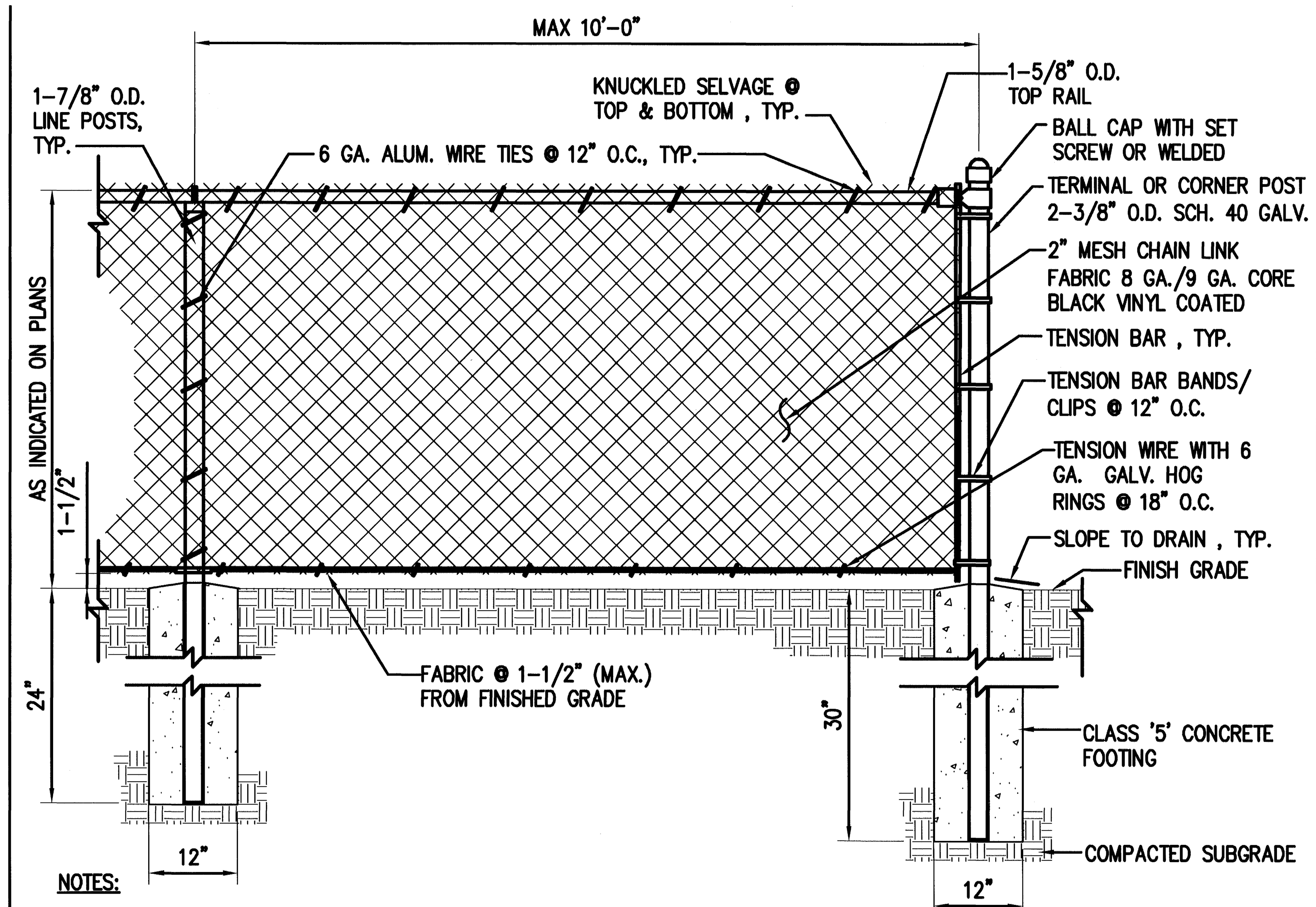


TYPICAL PEDESTRIAN GATE

NOTES:

1. FENCE FABRIC SHALL BE SECURED TO GATE FRAMES WITH KNUCKLED SELVAGE ALONG ALL EDGES FOR ALL TYPES CHAIN LINK FENCE INSTALLATIONS.
2. REFER TO SPECIFICATIONS FOR ALL PIPE SIZING, WIRE FABRIC GAUGES, FOOTING SIZES & OTHER APPLICABLE INFORMATION.
3. ALL CONCRETE FOOTINGS AROUND POSTS AT GROUND LINE SHALL BE MOUNDED & SLOPED FOR POSITIVE DRAINAGE.

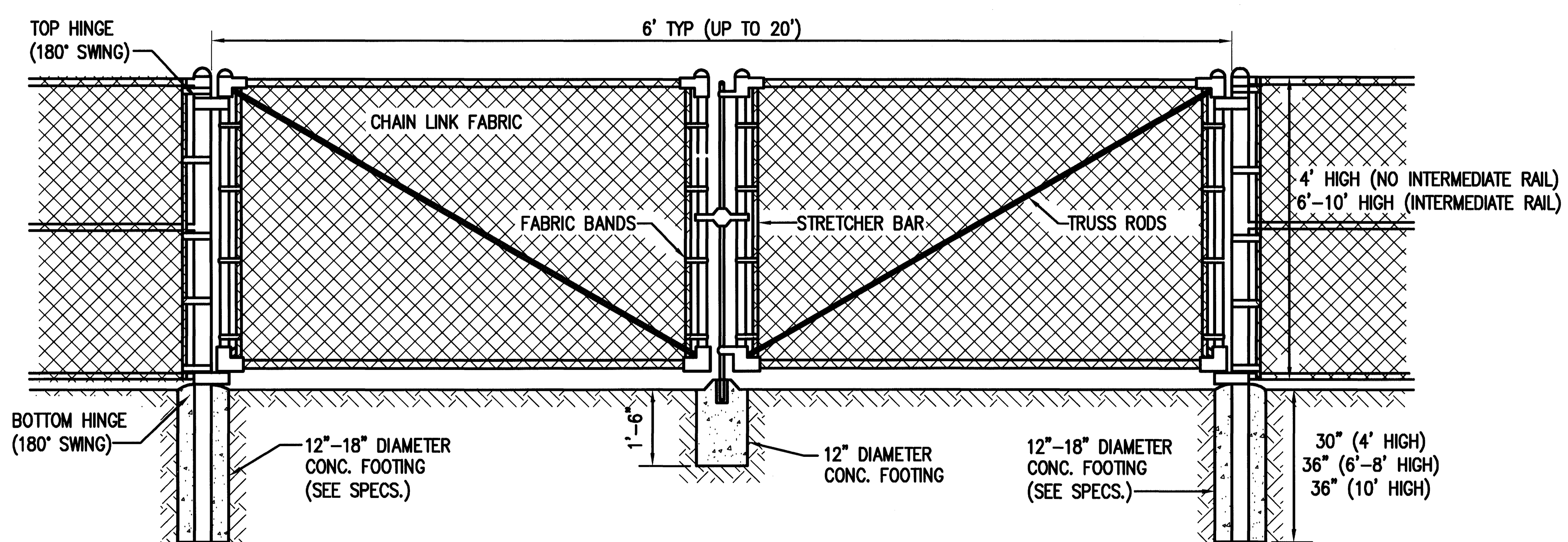
1 C302 SINGLE GATE DETAILS
SCALE: NTS



NOTES:

1. FIELD VERIFY LOCATION OF GATE AND CENTER SIDEWALK ON GATE ENTRANCE.
2. SEE SPECIFICATIONS FOR ALL APPLICABLE INFORMATION.
3. BOLT THREADS AND WIRE TIES FOR FENCING SHALL EXTEND AWAY FROM THE PLAYGROUND (OUTSIDE). FOR FENCING BETWEEN PLAYGROUNDS, BOLTS WIRETIES, ANY SHARP EDGES, SHALL BE GROUND SMOOTH OR CAPPED OR CAULKED.
4. SEE LANDSCAPE DETAILS FOR COMBINING MOW STRIPS AND FENCE POST FOOTINGS.

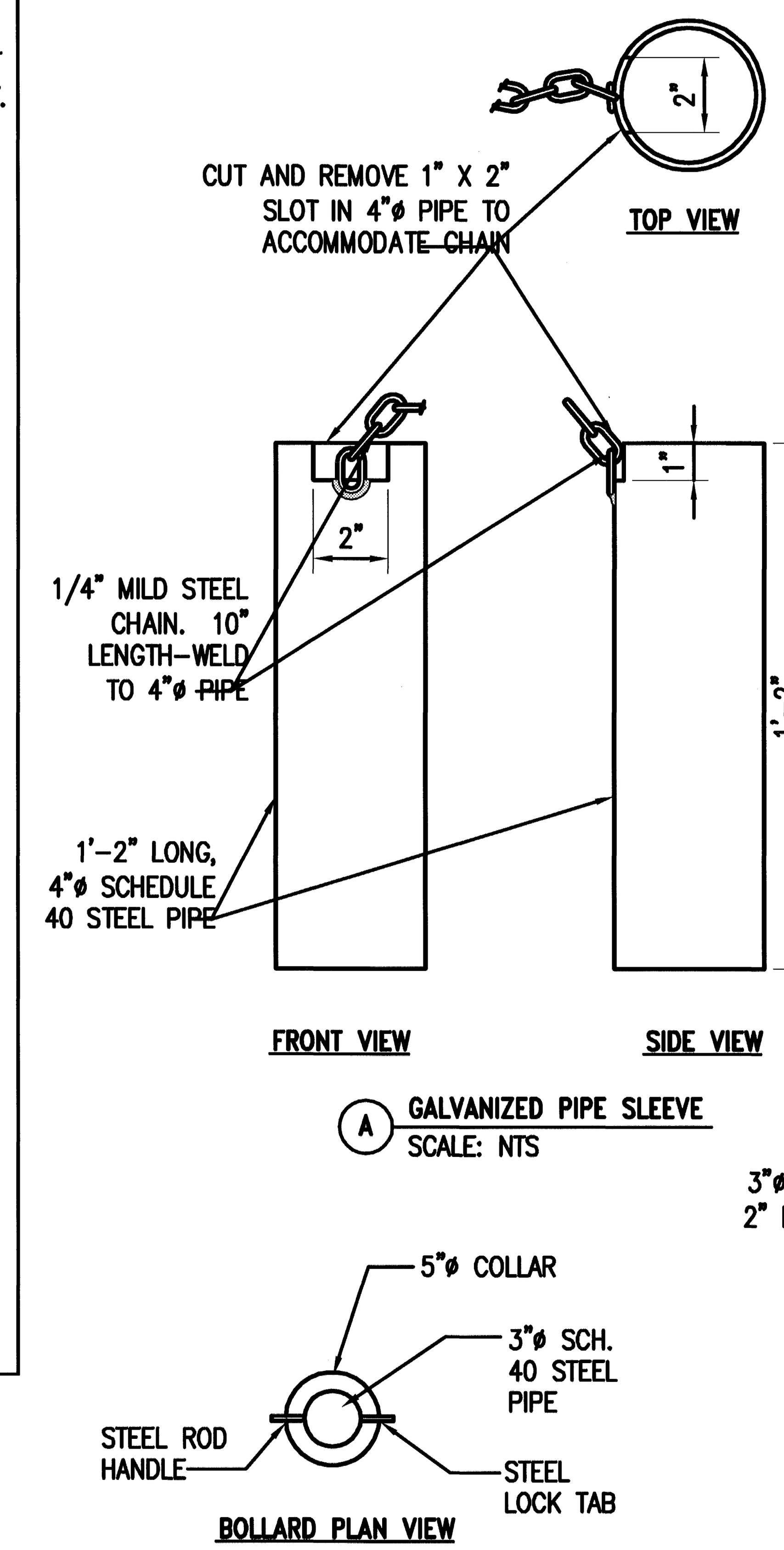
2 C302 CHAINLINK FENCE DETAIL
SCALE: NTS

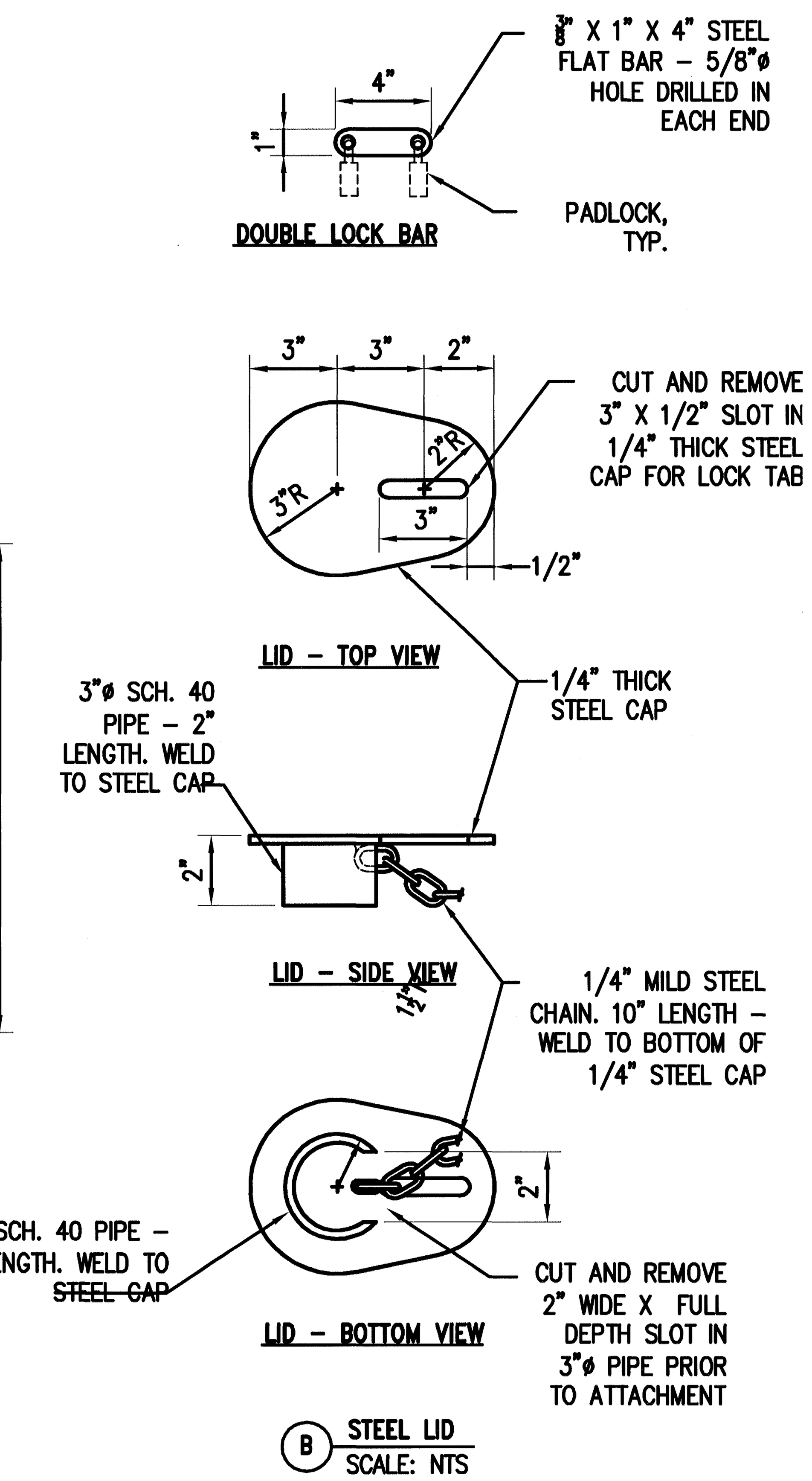


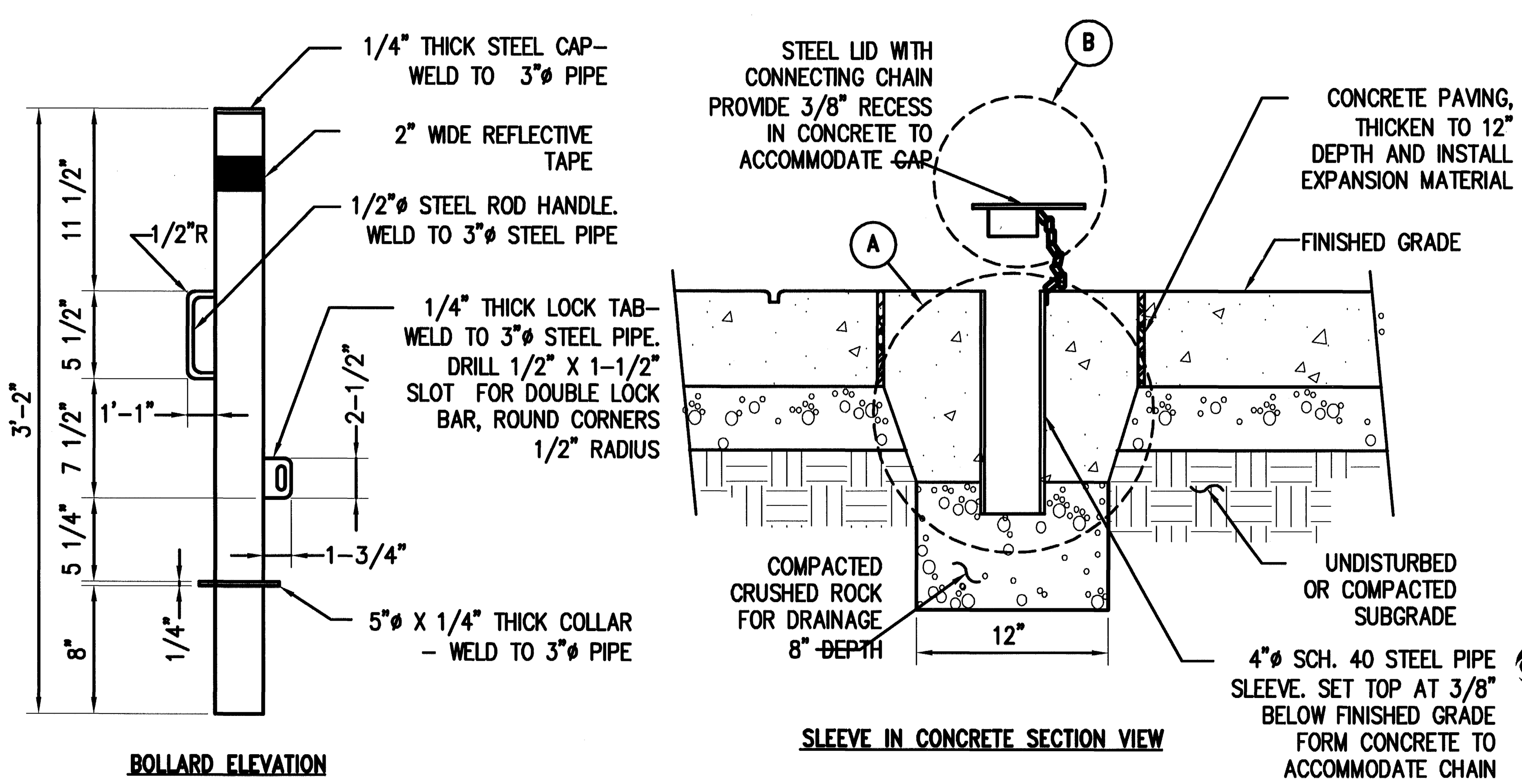
NOTES:

1. FENCE FABRIC SHALL BE SECURED TO GATE FRAMES WITH KNUCKLED SELVAGE ALONG ALL EDGES FOR ALL TYPES CHAIN LINK FENCE INSTALLATIONS.
2. REFER TO SPECIFICATIONS FOR ALL PIPE SIZING, WIRE FABRIC GAUGES, FOOTING SIZES & OTHER APPLICABLE INFORMATION.
3. ALL CONCRETE FOOTINGS AROUND POSTS AT GROUND LINE SHALL BE MOUNDED & SLOPED FOR POSITIVE DRAINAGE.

3 C302 TYPICAL DOUBLE GATE
SCALE: NTS







NOTE:
ALL PIPE SECTIONS SHALL CONSTRUCTED OF SCHEDULE 40 STEEL PIPE AND ALL COMPONENTS SHALL BE HOT DIPPED GALVANIZED AFTER FABRICATION.

4 C302 REMOVABLE STEEL BOLLARD DETAILS
SCALE: NTS

AS BUILT DRAWINGS

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US Army Corps of Engineers
Seattle District

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Drawn by: DLY	Design file no.
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Submitted by:	Section
Chief,	

U.S. ARMY ENGINEER DISTRICT, SEATTLE
CORPS OF ENGINEERS
SEATTLE, WASHINGTON

Submitted by: BergerABAM (206) 431-2300 FAX (206) 431-2250
Principal

FY09 ADDITION/RENOVATION
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Sheet 16 of 212